Mercedes-Benz

Guideline

Robot general

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Update service

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Change Journal

No	Chapter	Change content	Version	Date	Author
0		Adoption of document from Integra Integra wave 6/V60.07	70.00	2021-03	T. Breitmaier
1	All	• General adaptation of leasson learned • EZ image for K-pipe inserted • Insert construction site characters, due to clarification with collision and Tool64 • Technology packages-Description extended • Masref_User adapted	70.01	14.06.21	T. Breitmaier
2	All	• General revision of GL due to initial experience with KRC5. • Change is all marked yellow, but the deleted content is directly removed for better clarity! • => simple search with search term "V_70.02"	70.02	15.03.22	B. Gall; T. Breitmaier
3	4.4.2.1 5.4.10.2 5.5.7 5.4.9 3.15 Alle	• Cleanup of the main programs • No acceptance by Kuka for roller hemming • Load case measurement from 10 kg to 5% • Adjusted reference to interlocks • Update Reference trajectory • Further general revision of the GL with SAK robot => simple search with search term "V_70.03"	70.03	12.12.22	B. Gall
4	1.8.1 1.8.4.4 1.8.4.5 2.4.6.1 4.4.2.1 5.3.2 5.4.9 6.2.1	• general robot controller (external hard drive …) • stacking of cabinet only with approval • 600mm space on right side of cabinet • Kuka-Load-Online as new load check tool • main program for stud welding • commands use in programs (synout, comments, …) • extension for locks • - Kuka-Load-Online for load data documentation	70.04	29.03.23	B. Gall
5	All	Further general revision of the GL with SAK robot => simple search with search term "V_70.05"	70.05	07.12.23	B. Gall
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The current changes compared to the older version are highlighted in yellow.

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Foreword

The sum of all **available guidelines GL-DE** (**German**) and **GL-EN** (**English**) and **material release** lists (**MFL**) forms the basis for the comprehensive standardization of all technical scopes for production and plant technology at Mercedes-Benz AG.

Details for which precise information is not given in this guideline, or also points that are not covered in the associated electrical or mechanical standard documentation and in the corresponding software standard, shall be clarified with the person responsible at Mercedes-Benz AG.

Proposals based on the existing guidelines, hardware standard documentation and software standards shall be developed by the supplier and presented to the person responsible at Mercedes-Benz AG for approval.

The supplier is obliged to inform Mercedes-Benz AG of any necessary improvements, modifications and additions to this guideline.

Approved improvements, changes and additions will be integrated into a subsequent version of this guideline by Mercedes-Benz AG when updating the documentation.

This guideline doesn't release the supplier from its responsibility for the proper design of the systems and documentation to be supplied, taking into account the laws, regulations, conditions and requirements that apply at the respective production site.

In the event of contradictions in the guidelines or invitations to tender, the more stringent requirement shall apply.

Document content

General Robot Technology Guideline

Info:

- Country language points are marked in light blue!

The identification in the square brackets behind the chapter heading indicates when the last change took place **in this chapter.**

Example	Description
Chapter heading[V_70.00_iX]	Content identical to in integra phase X
Chapter heading[V_70.xx]	Content from the current wave/version

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1 General

1.1 Introduction [V_70.05]

This guideline was created with the aim of showing the contractor (hereinafter referred to as contractor) the concepts and systems defined by Mercedes-Benz AG (hereinafter referred to as client), as well as the functions of the robot and its peripherals, and standardizing them within the project or MBAG.

It also serves as a calculation aid for the underlying order. However, it doesn't release the contractor from its responsibility for achieving the described functional performance and for compliance with all regulations, guidelines and generally accepted technical rules to be taken into consideration.

The scope of delivery includes all required services and deliveries for a functioning and complete implementation of the functions and requirements described in the individual chapters of this policy. If the provider believes that the specifications made in this request cannot fulfill flawless function within the meaning of the task, the provider is obliged to object.

If information is to be contradicted in the higher-level tender or this document, the procedure must be coordinated with the Mercedes-Benz Robot Project Team. The higher-quality requirement must always be taken into consideration for cost analysis.

Changes that arise during an order shall only be made after consultation with and written assignment by the Mercedes-Benz Robot Project Team or the contractor. Changes not coordinated with the Mercedes-Benz Robot Project Team are not recognized and therefore not remunerated. Any non-disclosure on our part of proposals, claims, or evidence shall in no case be considered consent. No acceptance takes place if the content described in this guideline is not complied with.

The contractor is responsible for ensuring that the current version of this guideline is always available to its sub-suppliers.

All chapters described in the guideline refer to the current KUKA KRC5 software release V 8. X.x. The robot system shall be installed by the contractor with the project sample plate created by MBAG. The introduction of newer software releases and technology packages can result in changes that require a new version of the guideline. The change effort due to software updates etc. Shall be included in the contractor's scope of services. The contractor's services required for this shall be performed by the contractor in coordination with the Mercedes-Benz Robot Project Team.

All persons assigned to the programming of the robot control system must have appropriate knowledge (e.g. Training confirmation).

Each contractor shall appoint a competent robot specialist to coordinate his project. This person forms the interface to MBAG for organizational and technical processes. Tasks include daily reporting, relationships between robots and peripherals, as well as project engineering and programming questions, etc.

An overview of training courses on robot technology can be found in "Volume 3 Chapter 008. Qualification Measures".

Kuka documentation can be accessed via Xpert.kuka.com.

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Material release lists (MFL) can be found in the DEEP system (Digital Equipment Engineering Portal).

1.2 Other applicable documents

1.2.1 Documentation in ECM Volume 3 Chapter 040 [V_70.02]

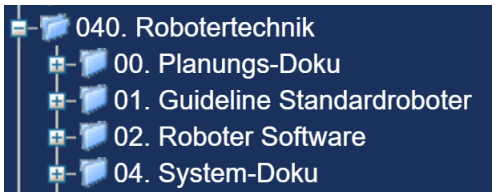

Figure 1 ECM overview

1.2.1.1 „00. Planning documentation "[V_70.02]

General information compact in a planner Guideline explains for standard robots and LBR.

1.2.1.2 „01. Guideline for Standard Robot "[V_70.02]

Standard guideline that generally applies to all shops.

Folder	Content
Folder 01	Basic GL robot Additional document for technology specific robot topics
Folder 02	General drawings for the robot.
Folder 03	EA table
Folder 04	Description of safe functionalities
Folder 09	Description of commissioning and kinematics parameterization as well as the backup/restore manager incl. Setup manager for the KSS software used
Folder 20	GL for the LBR robot

Table 1 ECM folder structure – Guideline standard robot

1.2.1.3 „02. Robot software "[V_70.02]

Contains the entire software (iKop) and its documentation of the standardized technology packages provided by MBAG.

1.2.1.4 „04. System documentation "[V_70.05]

Folder	Content
10. Kuka-Doku & Data Sheet	Find in Kuka Xpert (Xpert.kuka.com)
20. System-Dress-Package	Find in material release in DEEP

Table 2 ECM folder structure – System documentation

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1.2.2 Documentation in ECM Volume 3 Chapter 002.04.32 [V_70.02]

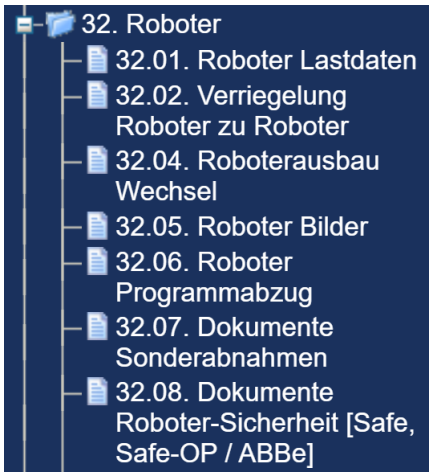

Figure 2 ECM overview of documentation

=> See Chapter 6.4 Acceptance [V_70.00].

1.2.3 Documentation in ECM Volume 3 further assigned chapters [V_70.00_I6]

1.2.3.1 Volume 3/008.02 Robot qualification [V_70.00_I6]

Information on robotics qualification measures.

1.2.3.2 Volume 3/further in sub-chapters [V_70.00_I6]

Further chapters such as E-plans, checklists and similar are described in the various known sub-chapters, but are not explicitly responsible for robotics!

1.3 Industrial Robot Safety [V_70.00_I5]

The robot system may only be used in technically perfect condition as well as intended for safety and danger. The use must take place in compliance with the existing operating instructions and specifications and the manufacturer's declaration enclosed with the robot system upon delivery. Malfunctions that may impair safety must be rectified immediately.

1.3.1 Smartpad operating modes-key [V_70.03]

According to documentation/operating instructions KUKA, the key must not remain on the SmartPad. The contractor shall remove the keys directly during initial startup and only qualified persons for robotics receive this corresponding to the risk assessment/definition.

When the system is handed over as per technical acceptance or BGB, the keys are handed over to a person which is defined from the robotic project team.

1.4 Approved types and options [V_70.02]

Approved robots are stored in the project order list.

a) See for Bodyshop volume 2/Chapter 226 Provision Process / robot provision

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b) See for paint shop volume 3/ chapter 40.340.01.20 robot provision
c) See for assembly volume 3/chapter 40.440.01.20 robot provision

1.5 General assembly specifications

1.5.1 Robot+peripherals general [V_70.03]

All robots may only be mounted on the floor in a standing position (0 degrees). If possible, the installation should be perpendicular to the main transport direction of the system. The robot shall be positioned in the transport position with aids coordinated by the Mercedes-Benz Robot Project Team (e.g. Fork-lift truck, mobile crane) in the system can be replaced without removing adjacent system components. The technical data of the aids shall be requested from the Mercedes-Benz Robot Project Team. Otherwise, the removability shall be ensured via steel construction. All deviations must be coordinated with the Mercedes-Benz Robot Project Team.

The exchange time of the robot mechanics is defined by the project in Volume 2.

All means of transport and tools required for replacement are part of the contractor's scope of delivery. Documentation of the replacement of the robots is according to the sample (Volume 3 Chapter 002.04.32.04. Robot removal concept) for the system acceptance. A layout shall also be created on which the removal situation (robot type, aids, location of aids, etc.) Is shown for each robot.

The design (main lines and secondary cells) must already be coordinated with MBAG in planning.

Furthermore, accessibility in the finished system for maintenance work (oil change, adjustment etc.) And tool change in the intended means of transport (e.g. Gun carriage). If this is not possible, the necessary aids (maintenance platform etc.) Are part of the contractor's scope of delivery. In addition to the linear unit, care shall be taken to ensure that a maintenance area for the carriages is always to be complied with at the end pieces according to Kuka documentation. Assembly space for plug fields etc. Shall be provided.

The attachment shall be designed according to the manufacturer's specifications Kuka documentation or other robot types. Accessibility of the screws and measuring tips for the purpose of assembly and disassembly shall be ensured.

If tools are used to initiate external forces on the robot system, they shall be taken into account in the KUKA load calculation of the robot and specified accordingly in the KUKA load powerpoint for the design release. In addition, special applications with force-acting processes such as roller hemming the Chapter 5.4.10.2 must be observed.

1.5.2 Robot Mechanics/Peripherals [V_70.03]

All robot tools with the exception of automatic exchangeable devices and measuring robots shall be installed and pinned on axis 6 with flange-mounted jigs and fixtures according to MBAG standards.

For more information on the flange connection and robot flange plate, see GL "000_B3.040.01.01_ GL-EN_Robot_Supplement_Vxx.xx".

The specifications of the robot manufacturer must be observed for the assembly of tools on the robot flange (e.g. screw length, screw quality, ...).

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1.5.3 Robot alignments incl. Linear unit [V_70.00_I6]

See also Chapter 2.4 Location and determination of tool data and coordinate systems.

1.5.3.1 Robot on Linear Axes KUKA [V_70.00_I6]

The coordinate systems on the linear unit are implemented according to the KUKA standard!

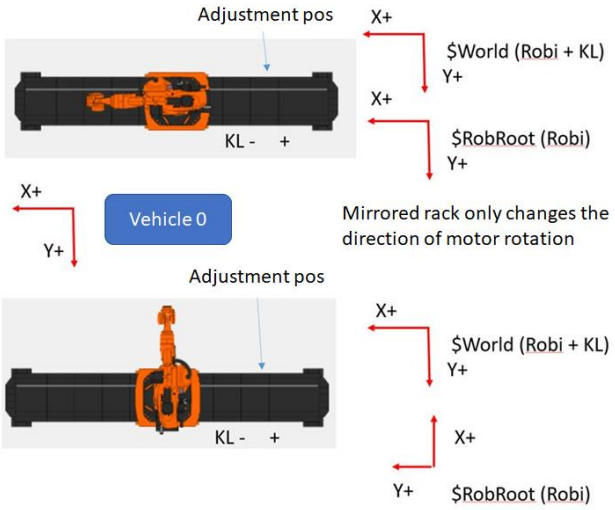

Figure 3 Orientation of KUKA linear unit

Means that no matter which direction the robot has on the linear unit, the $World is always identical!

1.5.3.2 Robot on linear axes in the surface [V_70.02]

Linear unit from other manufacturers is also used in the paint shop, as the KUKA linear unit is not suitable for all ambient requirements!

In addition, it must be noted that the coordinate system on the linear unit is different in the surface. See additional GL Volume 3 Chapter 040.340.01.10_GL-DE_Roboter-Erw.OF

1.5.3.3 Alignment to body in assembly [V_70.02]

Coordinate system for the KUKA linear units shall be implemented according to KUKA standard.

The alignment to the vehicle/body shall always be carried out in the 90° grid. All other orientations are not permissible!

See additional GL Volume 3 Chapter 040.440.01.10_GL-DE_Roboter-Erw.Assembly

~~#### 1.5.3.4 Robot with rotating table/rotary drum [V_70.06]~~

~~The Coordinate system for KUKA positioners shall be implemented according to KUKA standard.~~

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Figure 4 Alignment of rotating table KP1-V1000

1.5.4 Robot linear unit/peripherals [V_70.06]

The robots may only be mounted in 90° grid on the linear units. The grid of 45° is not permitted.

Space for changing the rollers shall be provided in accordance with KUKA documentation on the front faces of the linear unit.

All deviations must be coordinated with the Mercedes-Benz Robot Project Team.

The Z height is always set to 0mm.

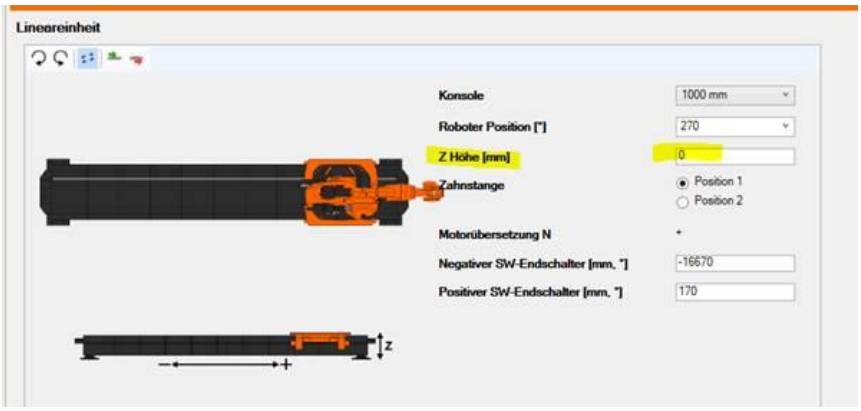

Figure 5 Z-height (0mm) linear unit

The linear unit shall always be defined as axis 7/E1.

In addition, the default setting "Use dynamic model" is always to be activated in the configuration.

The following options are possible:

- Mounting types:
  - Standard: Foundation plates for concrete floor
  - Floating foundation plates for expansion joints-Crossings=> Note that this is correspondingly wider and therefore the total KL is wider with cable drag
  - Steel construction: For assembly on steel substructures
  - Heavy duty foundations for Fortec robots with console

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    - Info on uneven floor conditions: If the compensating rings of the KUKA spindles are not sufficient, a steel structure may be installed underneath similar to the floating foundation plates. The plant manufacturer is responsible for this!
- Lubrication unit=> is included in the standard
    - Info: Activation of the lubrication unit shall be changed by AN depending on the mileage. If the lubrication unit is to go empty until the technical release, the contractor shall replace it!
- Consoles height
    - As per manufacturer's specifications
- Cable drag versions:
    - Exposed cables between the connection plate and the entrance of the drag chain must be protected against welding spatter and step on from the contractor.
    - Right and left of the axle
    - Internal cable trolley=> release, see Volume 2/Robot provision
        - Only suitable for handling application
        - Only up to a length of 8.4m stroke possible!
    - Info:
        - Cable drag lines must be kept free over the entire linear length (no obstacles)!
        - The cable drag line protrudes beyond the linear length on the roll side. See KUKA documentation
- ~~High-speed variant=> only for special approval by robotics project team~~
- Covered version is standard

1.5.4.1 Acceptance of Linear Unit [V_70.06]

Information from scope of supply, Volume 2/Robot provision

The scope of supply and services shall be designed according to the KUKA specification.

Leveling of leveling by a measurement report!

Acceptance of KUKA linear unit (optionally by KUKA)

Description of performance:

- ⇨ According to manufacturer specificationst
- ⇨ Check installation according to layout
- ⇨ Robot position on the carriage
- ~~⇨ Leveling of leveling (+ / - 0.1mm per meter)~~
- ~~⇨ Transitions of rails and racks~~
- ~~⇨ Check rolling behavior and assembly of cable drag~~
- ~~⇨ Installation of wiring to robot~~
- ~~⇨ Creation of the acceptance protocol (measurement protocol in tabular form with the leveling values per meter)~~

1.5.5 Robot rotating table/peripherals [V_70.06]

~~As rotating table only the KP1-V1000 HW-2 is released for the technology Perceptron inline measurement as special usage.~~

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Rotating tables see in “000_B3.027_GL-EN_Gerneral_Turntables Vxx.xx.pdf”.

1.6 Hose packages

1.6.1 General [V_70.05]

Robot energy supplies are required for supplying electrical, hydraulic and pneumatic consumers to the robot tool. Depending on the robot type used and the technology, they can be designed as an external or internal package.

If a standard hose packages from KUKA is ordered, this is always delivered assembled. Assembly shall only be performed by the contractor if it is special hose packages. In principle, the contractor is responsible for the proper execution of the overall package according to the manufacturer's specifications. This applies in particular even if he commissions subcontractors with the optimization service.

Special packages are not provided by Mercedes-Benz and are therefore part of the contractor's scope of delivery.

The contractor is liable for defects due to incorrect hose package installations or unfavorable movement programming. (See Volume 2 Provision Scopes Mercedes-Benz)

Standard hose packages consist of following 2 pieces:

- Axle 1-3
- Axis 3-6

Contents of the hose package shall consist of torsional-resistant cables, hoses, etc. => Drag chain suitability is NOT sufficient!

1.6.1.1 Axis 1-3 [V_70.00_I5]

The axle 1-3 package is designed as an internal package (hose routing carried out in the robot).

For some hose packages axis 1-3, a solution can also be run along the robot. This is specially designed in the hose package description.

Please note that the larger space requirement for the robot base in the system layout must be observed!

1.6.1.2 Axis 3-6 [V_70.00]

On axis 3 it is possible by adding additional fastening such as. Adhesive hose holder or impact assembly plate (not included in Mercedes-Benz scope of delivery) additional elements installed. Consultation in the regular communication Robotics project for approval

- ⇨ Unused contents in the hose packages must NOT be removed.
- ⇨ The unused contents are withdrawn to axis 3 so that the connectors on the bracket axis 6 are applied. Other hoses and cables are attached to axis 3 on the K-pipe bracket.

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1.6.2 Order [V_70.00_I5]

The standard energy supplies are ordered by the contractor as part of the robot order via the order list. For each robot, the desired energy supply variant shall be specified in accordance with the approval list when ordering. Incurred additional costs caused by the contractor in the order process (deadlines, design, variants, etc.) Shall be borne by the contractor. Mercedes-Benz must be informed immediately of any ordering process.

1.6.3 Commissioning [V_70.00_I5]

The commissioning of energy supplies takes place with the assembly of the corresponding robot tool.

Energy supplies shall be adapted depending on the robot's motion program and the arm extension used in the axis 3-6 part. The assembly specifications of the hose package supplier and, in particular, the following points shall be observed:

The hose package adaptation shall be carried out simultaneously with online programming.

No overload, crushing, and chafing on the hose package shall occur for all rail programs.

After completion of the movement programming, the hose guide shall be tested and adapted accordingly with regard to maximum permissible loads.

This includes:

- Permissible bending radii
- Tensile forces occurring
- Friction wear on interference edges or housing of robot
- Kinks especially in the area of the central hand
- Attachment of wear rings

Additional accessories (spring base, protectors, balancers, etc.) Required for the proper design of the energy supply shall be provided in the contractor's scope of delivery. The hose package A3-6 shall be exchangeable within 15min.

1.6.4 Acceptance [V_70.00_I6]

The contractor can have the hose package manufacturer carry out an acceptance test.

1.6.5 Project manager [V_70.00_I5]

The contractor appoints a person who is responsible for all aspects of energy supply within the scope of the project. This person is also the contact person for any problems that arise with the client.

1.6.6 Warranty [V_70.00_I5]

For damage in the period following provision by the client and acceptance (full capacity production), the contractor shall be liable.

For the hose package, the contractor grants a guarantee of 2 years from the moment reaching full capacity production, as described separately in Volume 2, in accordance with the scope of delivery of the robot.

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For all special hose packages, the contractor also shall guarantee a warranty of 2 years from reaching full capacity production.

1.6.7 Replacement parts [V_70.00_I5]

The contractor shall ensure replacement parts supply until acceptance.

1.6.8 Documentation scope [V_70.00_I5]

Digital photos of the robot system according to guideline shall be defined for final acceptance.

Furthermore, the contractor shall supply the complete technical documentation including bills of materials and the assembly specification for commissioning at the client.

1.6.9 Formats [V_70.00_I6]

All images are saved as JPG files. The documentation shall be supplied as part of the robot documentation. => see Chapter 6.4 Acceptance [V_70.00].

1.6.10 Technical Approval of Special Hose Packages [V_70.00_I5]

For all special hose packages (e.g. Adhesive bonding, clinching, bolt-, laser welding, etc.) Shall be mapped to a sample structure with functional peripherals by the contractor. In these packages, the standard components already used must be used as far as possible.

Each hose package variant shall be approved by the robot project team from the client.

1.6.11 Special Hose Packages [V_70.00_I5]

All hose packages always have a connecting point on the installation plate in front of the robot and in the height of axis A3.

1.7 Safety technology [V_70.06]

The standardization ISO 10218 Part 2 shall be implemented.

~~The SRM and SOP software cannot be installed simultaneously on the robot system!~~

~~1.7.1 Safe Range Monitoring "SRM" [V_70.06]~~

~~The SRM software from KUKA is included in the client basic scope of the robot.~~

~~Description of function is in Volume 3 Chapter 040.01.04 and checklist for acceptance in Volume 3 Chapter 002.04.32.08~~

~~Specifications for the risks to occupational safety in “Volume 3 Chapter 007” are binding.~~

~~The KUKA option "SRM" is approved as a function of the client.~~

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1.7.2 Safe operation "SOP" [V_70.02]

The safe operation functionality is described in Volume 3 Chapter 040.01.04 and checklist for acceptance in Volume 3 Chapter 002.04.32.08.

Specifications for the risks to occupational safety in “Volume 3 Chapter 007” are binding.

The KUKA option "SOP" is approved as a function of the client.

1.7.3 Safety axis limits [V_70.06]

When installing axle range limits (mechanical components), it shall be ensured that these are only approached in an emergency as this causes damage to the robot system. Inadvertent start-off in manual mode must therefore be prevented=> Specification of this for active SOP ~~or SRM~~ with a monitored area for specific axis!

If the robot enters this limit in an emergency, this shall be checked by the MBAG Robotics department or KUKA as to whether the robot mechanism may continue to operate!

Specifications for the risks to occupational safety in “Volume 3 Chapter 007” are binding.

The option is only approved as a special function from the client.

1.7.3.1 Brake test [V_70.05]

If the safe function is deactivated, the brake test function can be switched off according to the manufacturer's specifications. For this purpose, the system manufacturer shall define and determine the risks that are then posed by the robot system in a corresponding risk analysis. This consideration shall be submitted to the client!

Note: Make sure that the brake test is always carried out with the same weights (tool+additional loads). The advice is also that during tool change the brake test is carried out without tools.

The brake test must be switched off for additional axles that have no influence on overall safety, e.g. rotatable adhesive axles during assembly.

1.7.4 Fluid system (integrated in robot) [V_70.00_I6]

The installed fluid technology in the robot arm of the KUKA robot type Agilus may not be used at the client. The functions are not designed according to fluid standardization and GL Volume 3!

1.7.5 Safety connection [V_70.00_I5]

1.7.5.1 ProfiNETsafe [V_70.06]

The safety signals from the PLC to the robot are connected via the safety bus (Siemens ProfiNETsafe). The following documents are for information only. The configuration and programming is carried out by the programmer responsible for the safety PLC.

All robot-specific safety signals as well as the Safe-Operation ~~or Safe-Range Monitoring~~ function are included via the profisafe interface in the robot. The setting of the profisafe addresses must still be carried out by the contractor.

The corresponding description is included in the “DAI_Baseproject” tech package.

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No changes may be made to the wiring in the robot cabinet!

The Profisafe connection may only be used with the removed Siemens modules of the Integra project! (See fail-safe modules control technology)

1.7.6 Fluid control valve [V_70.03]

If there is a control air valve, this is controlled by an output from the robot controller.

In terms of safety, the control air valve is switched off via the US2 voltage.

1.8 Robot control system

1.8.1 Robot control system in general [V_70.04]

The KR C5 cabinet should be installed as close as possible to the robot. If combining robot controllers in a cabinet leads to longer connection cables of 15m, then the controllers must be accommodated in individual cabinets. Exception, if the robot cabinets have to be set up next to each other anyway due to the space available at the cells, then they can be combined in one cabinet despite longer connecting cables.

⇨ Basically, the most economical variant should be implemented.

The installation instructions specified by the manufacturer, minimum distances, etc. Shall be observed.

The KRC5-Steuerung corresponds to the KUKA standard and therefore the specifications from the KUKA documentation apply.

The installation concept must be coordinated with the project team.

The controllers in a KR C5 cabinet must always be assigned to the same PLC area.

An external hard drive is supplied with each controller in the basic configuration, with which the robot must be commissioned and operated.

1.8.2 Cabinet integration system [V_70.00]

Control cabinet KRC 5=> see electrical diagram

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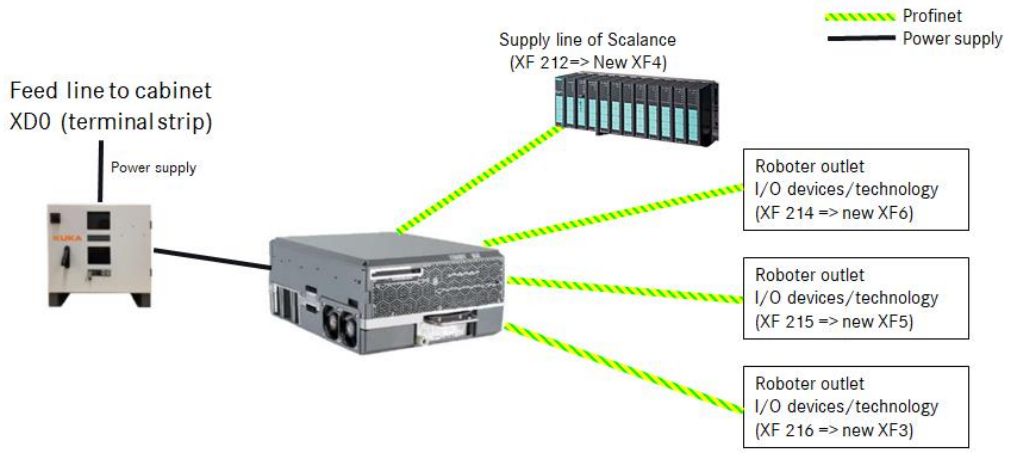

Figure 6 Cabinet connections

1.8.2.1 Connection options [V_70.00]

Standard:

4x Profinet, see Chapter 1.8.3.1 Profinet connection [V_70.00]

1x SmartPad interface

1x RDC interface

1x safe adjustment test

Options:

1x RoboTeam IN+Out (RoboTeam option)

1x fast measurement (option s-measurement)

Cabinet options shall be specified by the contractor in the robot order list. Subsequent installation by KUKA is at the expense of the contractor.

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1.8.3 Mercedes-Benz Profinet

1.8.3.1 Profinet connection [V_70.00]

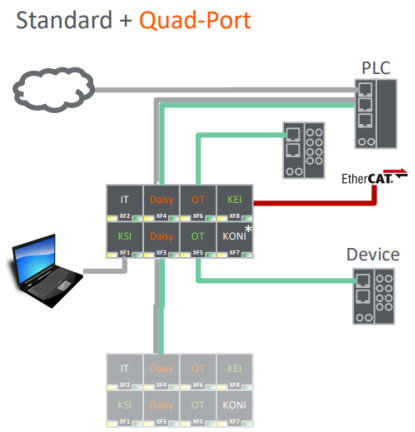

Figure 7 Network interfaces of controller

XF4 Profinet IN

XF5 Profinet OUT I/O devices/technology

XF6 Profinet OUT I/O devices/technology

XF3 Profinet OUT I/O devices/technology

1.8.4 Cabinet Installation

1.8.4.1 Installation of handheld programming device [V_70.05]

It is intended that the handheld programming device (SmartPad) are mounted as close as possible to the access point of the system, e.g. Next to the safety door. Installation location should be protected from direct flying sparks. Cable lengths of the handheld programming device must be designed in such a way that it can be used in the entire working area of the robot.

Cabinet size	SmartPad assembly at Cabinet allowed
2s	No. Exception: If the cabinet has at least a distance of 200mm to the ground.
3s	Yes.

Table 3 SmartPad assembly on cabinet

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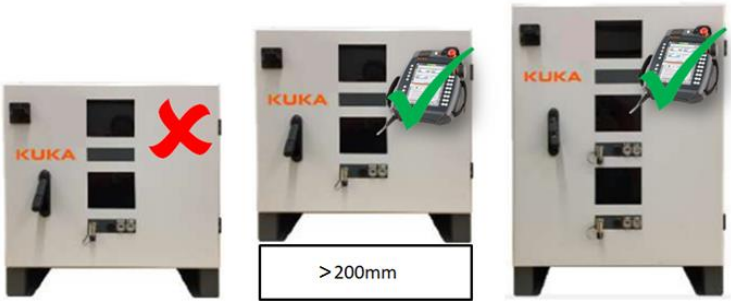

Figure 8 SmartPad assembly on cabinet

The installation of max. One programming hand (SmartPad) is possible at a cabinet.

Smartpad extensions are possible according to manufacturer specifications. If Smart-Pad cables are extended or a cable reel is used, the connector must be mounted with a clamp outside the cable tray.

Figure 9 Attachment for SmartPad cable connector

Cable winders from Sumcab type "MTPReel MED" may also be used. Sumcab offers mounting options on the robot cabinet or on the protective fence. (Is not provided)

⇨ Practical application location shall be coordinated with project team. Universal use is not planned.

Figure 10 Smartpad cable reel

1.8.4.2 Cable reserve winding [V_70.02]

The connection cable length to the robot shall not exceed the lengths defined in the project order list. In the case of a KRC5-Steuerungstausch the cabinet no longer needs to be exchanged as before with the KRC4-Steuerungsgeneratund and therefore the cable reserve is omitted.

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The cables may not be routed in a stramm, i.e. No cable on the cable to load the cable entry into the cabinet cabinet.

If reserve is still available during the cable selection , it must be stowed properly on the cabinet side.

Nothing can be mounted on the cabinet itself.

1.8.4.3 KUKA Cabinet/stack for technology [V_70.03]

Other cabinets may be mounted on the KUKA cabinet! Assembly is possible directly without body frames or similar. In the case of direct assembly of the KRC5 cabinet and technology control, the contractor is responsible for a constructive adaptation.

Pay attention to the degrees of freedom/accessibility of the cabinet (cable insertions at the rear, right/front doors).

Please note the KUKA specification e.g. Load, heat development, etc.

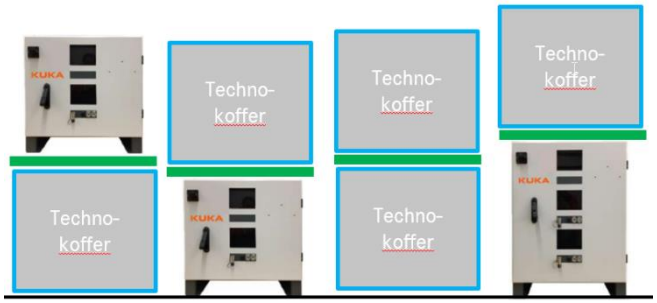

Figure 11 Robot cabinets stacked with technology

1.8.4.4 KUKA Cabinet/stack from 2x cabinets [70.04]

If KRC5 cabinets are to be stacked, the Kuka specifications must be observed. Direct screwing of the cabinets is permitted.

Stacking of KRC5 cabinets only with the approval of the robotics project team.

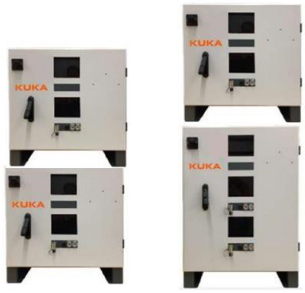

Figure 12 2 Robot cabinets on top of each other

1.8.4.5 Cabinet spacing [V_70.04]

Extended definition of Mercedes Benz AG:

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- Space in right side 600mm

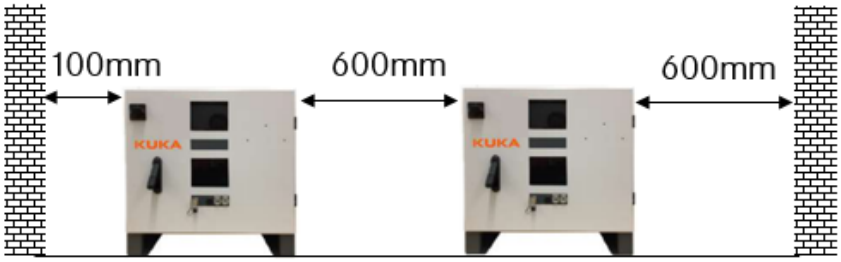

Figure 13 Minimum distance cabinets

1.8.5 Network connection [V_70.00]

All robot systems have an Ethernet interface according to IEEE 802.3 (100BaseT). This includes the following tasks in KRC5-System:

- Profinet connections
- Archiving
- Remote maintenance
- Access for work visual
- Etc.

The "Roboteam Geometric Cooperation" functions are used as an option on individual systems. For this purpose, the robots shall be equipped with a separate Ethernet interface. Details can be found in the separate guidelines.

1.9 Naming of robot system [V_70.02]

1.9.1 Robot name [V_70.02]

The robot name is assembled as follows:

< *System name* > _ < *Station* > *RB*_ < *Robot number* >

E.g. *HD61_110RB_100*

1.9.2 WorkVisual [V_70.02]

Is 4.1 Project structure WorkVisual described in Chapter.

⇨ Always includes robot and system names.

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1.9.3 Host name [V_70.02]

The host name for Windows is defined (unless otherwise defined in Volume 2) as follows:

Example					
Plant code	System	Station	Robot	Robot number	Description
1-4	5-8	9-11	12	13-15	Letter point
M050	HD61	110	R	100	Example: M050HD61110R100 If system name has 4 characters Robot 100 of station 110 at plant HD61 in plant 050

Table 4 List host name

If the system name has more than 4 characters, use the following scheme.

Example (more than 4 characters for the system					
Plant code	System	Station	Robot	Robot number	Description
1-4	5-9	10-12	13	14-15	Letter point
M050	BSZ01	110	R	10	Example 5 characters for system: Robot 100in station 110 at plant BSZ01 in plant 050.
1-4	5-10	11-13	14	15	Letter point
M050	M62409	110	R	1	Example 6 characters for system: Robot 100in station 110 at plant M62409im plant 050.

Table 5 List of hostname for more than 4 characters

1.9.4 PROFInet device name [V_70.02]

See also Chapter5.6.2.2Profinet naming [V_70.03].

The PROFInet device name of the robot, as well as the PROFInet names of the subordinate devices, must be consistent with the general description of the control engineering description in Chapter B3-026-03-01 (000_B3.026.03.01_GL-DE_HWKonfig_PROFINET_VXX.XX).

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2 Commissioning

2.1 Labeling [V_70.03]

2.1.1 Labeling KR C5 Controller and Cabinet [V_ 70.03]

See "Volume 3 / 004. Designations and markings" for labeling specifications.

2.1.2 SmartPad lettering [V_ 70.05]

Labeling from SmartPad and SmartPad holder is not permitted! ~~No labeling permitted!~~

2.1.3 Labeling of Robot Mechanics [V_70.02]

The robot mechanical system shall be labeled at the front and rear on the axle-2 and shall be clearly visible from both sides. Example: 030RB_200

- Font size at least 50mm

Figure 14 Labeling of robot mechanical components (A2)

2.1.3.1 Safe Operation~~/Safe-Range Monitoring~~ Active Labeling [V_70.06]

If Safe Operation ~~or Safe-Range Monitoring~~ is activated on the robot system, a black "S" on both sides of the axle-2 shall be attached as additional labeling on the yellow ground.

Figure 15 Labeling of SOP

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2.1.3.2 Safe axle range limitation Active labeling [V_70.02]

If safe axis range limits have been mounted on the robot system , a black "H" on yellow ground shall be installed on both sides of the axis 2 as additional labeling.

2.1.4 Labeling of reference switches for SOP ~~/SRM~~ [V_70.06]

The reference switches for the Safe Operation ~~and Safe-Range Monitoring~~ software shall be labeled with the associated robot name.

- ⇨ Aluminum shield (see “000_B3.004.02_GL-EN_Labeling_Vxx.xx”)
  - Size 15x50mm

2.2 Floor condition [V_70.00_I5]

In principle, the requirements of the robot manufacturer regarding the floor condition shall be taken into account. If the floor doesn't meet these requirements, the contractor shall immediately report this to Mercedes-Benz. In no case shall the robot be mounted in these cases, without the express approval of MBAG.

The robot shall not be mounted on expansion joints. This also applies to base assembly. There shall also be no expansion joints between robot and its work and system stations. The contractor shall examine this in advance and take it into account in layout design.

2.2.1 Anchoring Concept[V_70.05]

See GL “000_B3.012_GL-EN_Floorclamping_Technology Vxx.xx”

2.2.2 Attachments for Robots and Linear Units [V_70.05]

The contractor shall provide written proof for the use and stability of the respective use case for the dowel attachments. The respective foundation fastening kit (4 individual plates) from Kuka shall be used for the robot installation (the foundation set is not an MBAG provision).

The foundation plate shall be leveled. A suitable compensating mass shall be used between the foundation plate and anchoring base if the permissible. Manufacturer tolerances of the anchoring ground are exceeded.

Furthermore, the system manufacturer can provide a large foundation plate for the robot installation. In this case, however, coordination with the robot manufacturer, dowel manufacturer, planner of the anchoring ground and the Mercedes-Benz robot project team is absolutely necessary.

The specifications of the robot manufacturer must be adhered to.

The tightening torques shall be applied according to the robot manufacturer.

2.3 Commissioning system [V_70.05]

Commissioning is carried out according to a separate specification.

See document Robotics in Vol. 2 and “000_B3.040.01.09_GL_EN_Robot_COM_KRC5_KSS87_V70.xx.pdf”.

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- ⇨ If the tracking tool "Coman" is used in the project, the content from the robot commissioning sheet is tracked with them and a signature MBAG/AG is no longer required on site!

The KUKA company commissions the entire robot mechanics and control cabinet system and saves the resulting drive bus configuration on the system. This pair must therefore be kept together by the contractor when it is set up, according to the manufacturer's declaration of incorporation. Otherwise, initial commissioning or the robot's procedure after switching on the operating voltage is not possible.

The contractor shall ensure that ordered robot mechanisms with their associated robot cabinets are carefully assembled.

A new plant commissioning by KUKA on site goes to cost-loads of the contractor.

The wiring of the robot controller takes place in the housing of the KRC5-Cabinets, in the KUKA specifications.

Incorrectly ordered cable lengths can only be exchanged at KUKA for additional costs to AN if the box has not yet been opened.

2.3.1 Robot adjustment [V_70.05]

Robot adjustment is necessary to compare the robot mechanism with the control "zero position ".

The starting point for all measurements on the industrial robot is the mechanical zero position of the axles. It is determined with an electronic measuring probe (EMD) for each axis and automatically approached during adjustment.

The system supplier must ensure that the environment allows EMD adjustment with the tool and hose package installed.

The robot adjustment is also simulated in the simulation (OLP).

The following procedure must be adhered to:

- ⇨ Before starting programming perform initial adjustment **without** tools and **without** additional load. For this purpose, the hose packages on the robot are not disassembled, which were delivered with the robot mechanism. “EMD – With load correction – First mastering”
- ⇨ Then mount the tool and the additional load and determine the difference to the initial adjustment with the "EMD – With load correction – Teach offset" function.
- ⇨ The " EMD – With load correction – Load mastering – With offset" function shall be performed for checking/checking with tool

During the tool learning drive, it shall be ensured that they always take place from the same starting position (pre-adjustment position of all 6 axes), as the forces acting on the engine depend on the current IR position. If the start position is not possible in the pre-adjustment position, an alternative position must be sought. In such a case, coordination with the specialist department is necessary. The axis positions must be marked here.

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A tool learning drive shall be performed for all tools used as soon as they have different load data (in config.dat).

When using tool changers, a tool learning drive shall be carried out for each tool.

2.4 Location and determination of tool data and coordinate systems

2.4.1 General [V_70.00_I6]

The control concept provides for the following Cartesian coordinate systems:

System variable	Coordinate system
$WORLD	World coordinate system
$ROBROOT	Robot coordinate system
$BASE	Basis (workpiece) coordinate system
$TOOL	Tool coordinate system

Table 6 Coordinate systems

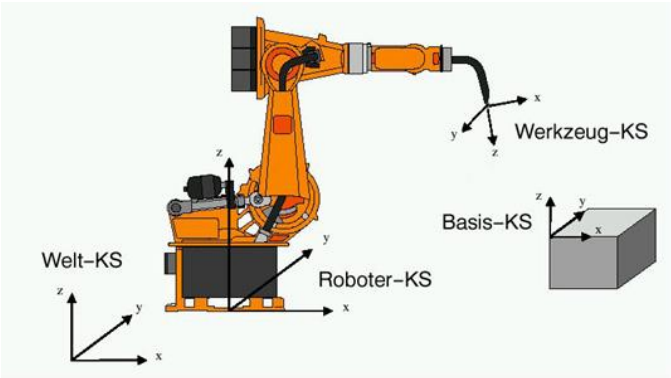

Figure 16 Robot coordinate system

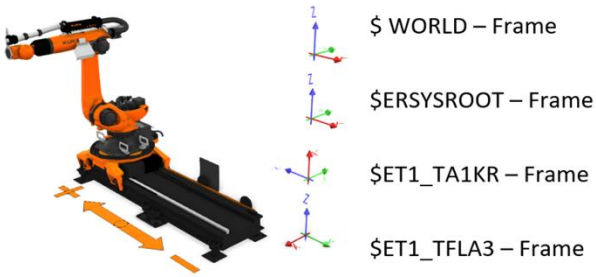

Figure 17 Robot coordinate system example of linear unit

2.4.1.1 World Coordinate System [V_70.00_I6]

The world coordinate system is a fixed (moving with a robot movement not) coordinate system, which serves as the origin coordinate system for a robot system (robot, component support or tool). It thus represents the reference system for both the robot system and the cell periphery.

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In the case of coordinate systems with a linear axis, ensure that the direction of travel of the linear unit and robot system matches!

2.4.1.2 Robot Coordinate System [V_70.00_I5]

This coordinate system is located in the robot base and serves as a reference coordinate system for the mechanical robot setup. It refers to the world coordinate system again and is identical with it upon delivery. With $ROBROOT, a shift of the robot to $WORLD can thus be defined. This coordinate system must not be changed.

2.4.1.3 Tool Coordinate System [V_70.00_I5]

The tool coordinate system has its origin in the tool tip. The orientation must be selected in such a way that its Z – axis is identical with the tool-thrust direction and points out of the tool. When the tool tip is moved, the tool coordinate system is moved along. During delivery, the tool coordinate system is in the robot flange (Z – axle is identical to axis 6). It refers to the robot coordinate system via the transformation.

If a tool change occurs, the original program can be reused after re-measurement because the computer's coordinates are known to the computer.

2.4.1.4 Basic coordinate system [V_70.00_I5]

The basic coordinate system is used as a reference system for describing the position of the workpiece. The robot is programmed in the basic coordinate system. It has the world coordinate system as a reference coordinate system. On delivery, $BASE=$ WORLD.

Due to a change of $BASE, several identical workpieces can be processed at different locations with the same program.

During the interpolation of the movement track, the robot control system normally calculates the current position ($POS_ACT) in relation to the $BASE coordinate system (tool on robot flange).

2.4.2 Determination Tool Geometry Data ($Tool) [V_70.06]

~~2.4.2.1 Determination [V_70.06]~~

For the motion planning of the robot control system, the position of the TCP (tool KS) shall be known in 6 degrees of freedom. This is determined with the so-called tool measurement, the scope of delivery of the contractor is or the data comes via a measurement.

The TCP measurement should be determined with the following priority:

1. OLP
2. TCP measurement on the robot (XYZ 4 - point method & orientation for ABC world 6D, see Kuka documentation)
3. External tool measurement

This shall be checked by the contractor on the construction site whether the TCP is correct! TCP must be checked with XYZ and orientation (ABC).

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Joining tools must be measured at the measuring points provided by the manufacturer. ~~The data must be entered in the Excel list provided.~~ This measurement report is supplied with the joining tool and must be handed over to the Mercedes-Benz robot project team.

The guideline “000_B3.040.01.01_GL-EN_Robot_Supplement_Vxxxx”, in which specific information is defined for almost every application tool, shall also be observed. ~~should also be observed, in which a definition for determining tool data is stored for almost every application tool.~~

Special tools are only permissible following consultation with the Mercedes-Benz Robot Project Team.

The position of the TOOL Center Point depends on the tool type used. A distinction is made between 2 cases:

(1) Tool with defined operating point (e.g. Welding gun, stud head)
(2) Surface tool without operating point (e.g. Gripper tools)

All of these tools “$Tool[x]” must be created by the contractor in the “Dai_Checktool”, auxiliary tools are not affected by this!

If tools correspond to case 1, the TOOL Center Point is ideally placed in the operating point.

Basically, for all tools, the direction of impact is pointing towards Z +.

~~Tools for which no measuring method is planned using a coordinate measuring machine shall be measured using the corresponding KUKA standard method, with a tool tip and specified operating point on the tool.~~

~~First, the translatory distances between the robot flange and the operating point of the tool are determined via the XYZ-4-point method. The orientation method in Worldcoordinates (ABC-World 6-D) measures the angles and the direction of impact. How the measurement program works can be found in the manufacturer's operating instructions.~~

The tool geometry data shall be entered before the start of the programming work at program release~~, as the robot control system stores all travel points in relation to the respective tool~~.

At stations where a replacement tool is planned, it must be replaced with the original tool before production test 2. The replacement tools must always be geometrically designed so that no program changes are necessary on the robot. The tolerance specifications shall be taken from the respective drawings; however, they shall not exceed 0.5mm.

~~2.4.2.1.1 Determination of tool geometry data ($Tool) [V_ 70.06]~~

~~See additional document of "000_B3.040.01.01_GL-EN_Robot_Supplement_Vxx.xx".~~

In the tools used, the online load data check must always be switched on with monitoring for "Warning" under overload or overload.

~~2.4.2.1.2 Overview of defined $tool 's [V_70.06]~~

~~See additional document of "000_B3.040.01.01_GL-EN_Robot_Supplement_Vxx.xx”.~~

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2.4.3 External tool ($Base) [V_70.00_I5]

External tool data is required if the robot leads the workpiece. This is the case for process grippers, for example, where the robot welds a component part in the gripper on a stationary spot welding gun. The operating point of this spot welding gun represents the external TCP, whose position must be measured for the robot's basic coordinate system.

The measurement is usually carried out analogous to the method described below using a coordinate measuring machine. Alternatively, the measurement can also be carried out with the measurement programs available in the robot control system. In this case, the 5D method shall be used by entering the rotation angle from the simulation.

⇨ See manufacturer documentation.

The measurement results shall be saved in the corresponding BASE field.

The position of the external TCP can be found in the following chapters.

2.4.4 External Robot Axes [V_70.06]

Only external axes that are coupled geometrically with the robot's procedure may still be connected.

Currently in use:

- KUKA linear units on which the robot mechanical system is mounted
  - KL4000-S (Speed) – Standard linear unit
  - KL5000 for heavy duty Fortec-2 ultra
- ~~KUKA rotating table (KP1) (Only approved for inline measurement-see GL-IMT)~~
- External rotating adhesive nozzle (assembly only)

External axes that are not geometrically coupled may only be moved when the robot is stationary.

These are then always actuated by the PLC or technology controls!

External axes must be calibrated specifically.

The assignment to create machine data for external kinematics is the scope of the Contractor. The following machine data are provided through WoV:

- Linear unit (KL)
- ~~Turn table (KP1)~~

See manufacturer documentation.

2.4.5 Basic coordinate system ($base)

2.4.5.1 General [V_70.06]

The overall concept "Digital Factory" is mandatory for robot simulation and offline programming (OLP). This means that the contractor shall model the entire system geometry in 3D, in which the layout of the robot cell is determined with all components in addition to functional analyses and many other parameters.

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As a result, a foundation plan is created, according to which all system components are positioned and fastened.

The robot is then calibrated using a coordinate measuring machine (tool/workpiece) and thus the real positions are determined. A data comparison must then be carried out in the OLP system.

Measurement shall be carried out with TOOL 63 with the expanding load data (incl. Measuring instruments for base measurement). This shall be reset to "0" after use.

Only measurement types are permissible

a. Enable comparison of measurement data with the OLP system
b. Do not install any additional software on the robot

In this process, the vector direction of the vehicle coordinate system (veh. 0 Frame) shall be determined for the robot coordinate system ($BASE) in 6 degrees of freedom.

Across the technology, the coordinates of the joints are compared between the TARGET data from the data record and the ACTUAL data reported back by the robot.
A deviation may be due to inaccurate base measurements. This must then be repeated if necessary!
The dimensional location tolerances of the joining technologies can be obtained from the respective MBAG departments in the projects.
A new base calibration shall be borne by the contractor. The system shall be operable again after remeasurement

2.4.5.2 Location and determination of vehicle 0 frames [V_70.00_I6]

To calibrate and check production systems, all component carriers such as devices must have the following features:

- Reference blocks with reference holes, diameter 1010H7, which are built on each unit and at the central measuring stations in direction for the grid position of the vehicle coordinate system.
- Directly next to these reference holes, the values (X, Y, Z depending on the vehicle position) are to be integrated for these so-called reference marks with number numbers.
- Compliance with the mechanical repeatability of < 0.4mm (6 sigma, total) of the component holders in all 3 coordinate directions.
- Swivel frames shall generally be mechanically locked.

Measuring features that are calibrated by the coordinate measuring machine are introduced in the holes and on the TCP of the robot tool during measurement. The vector $BASE is determined via the coordinate system chain robot tool and vehicle 0 frame. Alternatively, this may also be measured and determined via the robot bin. This offers the advantage that this can be done before the robot has been mounted. At least 3 points must be measured for this purpose.

Programming in the vehicle-0 frame brings advantages in offline and online programming, since the point coordinates are compatible with the specifications of the vehicle design. Furthermore, when re-programming the reference to the coordinate system, it is simplified as all components use the same system.

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2.4.5.3 Position Test Bases [V_70.00_I5]

Testing of bases can be realized in the process point in the device by switching to "Vehicle 0" data record in the $Tool / $Base.

2.4.6 Load data (tool and additional load)

2.4.6.1 General [V_70.04]

The tools Kuka-Load-Online (Kuka-Load) and LoadDataDetermination (LDD) are to be used for the load analysis.

Robot operation with entered load data is mandatory by the manufacturer. The operation with load data protects the robot mechanics as the acceleration profiles of the axles are calculated due to the entered load data and as a result, the transmission torques are reduced in the limit range.

The load data consist of the sizes weight, center of gravity distance and moments of inertia of the tool or additional load. On delivery, default data for tool and additional load are set in the control system.

When positioning the positioning of the IRs, the deflection is taken into account based on the load and additional load data entered, which directly influences the programmed positions. The load and additional load data must therefore be determined and entered before programming begins!

Make sure that the currently valid tool is entered in each PTP home.

For load changes e.g. Tool changers or component mounts must always be ensured that the home or transfer points are always defined with the correct load data. For tool changers, it may be necessary to define a second home. => Always coordinate in the project team in special cases.

2.4.6.2 Default values of the load data parameters [V_70.06]

This default values are already to be observed in the robot simulation and design!

Additional load data shall be entered as follows unless detailed specification is available in the Technology Guidelines.

	Technology without Handling (welding, riveting, ...)	Technology with Handling (e.g. BestFit, ...)
additional load	see K-Pipe „Welding“ in Kuka-Load	see K-Pipe „Handling“ in Kuka-Load
Axis 6 load	Tool weight + 13kg (for Cybertech + 3kg)	Tool weight + 7kg (for Cybertech + 3kg)
Max. workload	95%	90%

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If the limit values are exceeded, this robot must be approved in the Mercedes-Benz project team!

In the design phase, static or dynamic overload (red message) and warning (yellow message) are not permissible.

If double K pipe is used on axis 3, it must be entered in KUKA Load!

2.4.6.3 Determination of load data parameters [V_70.06]

The determination of the load data is divided into phases:

1. Design (by means of "Kuka-Load")
2. Weighing of additional load and complete tool with component
3. Load data determination on the robot with LDD (see manufacturer documentation)
4. Return to Kuka-Load for documentation
5. In case of borderline cases and small robot types, it may be necessary to evaluate the orientation values in Kuka-Load via the option and document the real data documentation.

All tool load data of the different phases shall be documented separately in the Kuka-Load & LDD for each system.

Once the tool has been designed, it and the load analysis created with Kuka Load must be submitted to the client's robotics project team for approval.

The specified values shall be taken into account when carrying out the load-bearing load of the robot tools.

In the case of additional loads (Axis1 - Axis3), if the standard periphery such as metering plates or valve batteries, the data is to be determined and entered once for each variant. In the case of non-standard peripherals, the additional load data shall be determined for each IR.

When determining the tool load data, make sure that the valid additional load data (Achse1-Achse3) is determined and entered in the control system beforehand. The determination shall also be carried out with the ready-to-use hose package.

Additional tools with adapted load data only need to be created if the component exceeds more than 5% of the robot's nominal payload, e.g. Quantec KR210 = 10.5 kg.

In the controller, the load data parameters are permanently assigned to the tool geometry data (TOOL_DATA [1] uses the load parameters LOAD_DATA [1] etc.).

The contractor must therefore ensure that if there are several load data sets, the associated tool geometry data (TOOL_DATA{XX]) is copied accordingly.

In addition, when programming it must be ensured that the movement points are programmed with the correct load data record. This case often occurs with gripper tools for which the robot performs movements with and without component.

The results of the load data determinations shall be entered in accordance with the documentation Volume 3 Chapter 002.03.32.01 and transferred to the Robot Project Team from the client.

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The load data determination shall be carried out by the contractor for each robot. In the auxiliary program dai_ldd_xx.SRC the travel distance from the HOME position to the start position shall be programmed. The work areas of the individual axes in the program must also be recorded as a comment.

Exceptions are always to be coordinated with the Robot Project Team from the client!

- In the case of very small tools that are too light for determining the load on the robot, the load data may be determined offline and entered manually into the robot.
- In the system it is not possible due to the space requirement that the robot LDD can make. In this case, either a protective fence can be dismantled or the load data can be determined outside on another robot and then entered into the affected robot.
- ~~If there are identical grippers including component weight within a system, then the load data can be copied to the identical robots.~~

If the LDD result is not successful, e.g. Dynamic overload/static overload/etc., This must be reported to the Robotics project team and change from the contractor.

In the tool itself, the online load data check must be activated with the assignment for "overload" and "underload" as a warning.

The contractor shall ensure that all tools and load data actually available are entered (including auxiliary tools). If tools are not used, the default values shall be used as tool load data.

As of LDD version 7.1, load data determination is also possible in T1. This MUST only be used for commissioning and must be repeated at a later time in T2. This is necessary so that the moments of inertia can be determined more precisely by the LDD!

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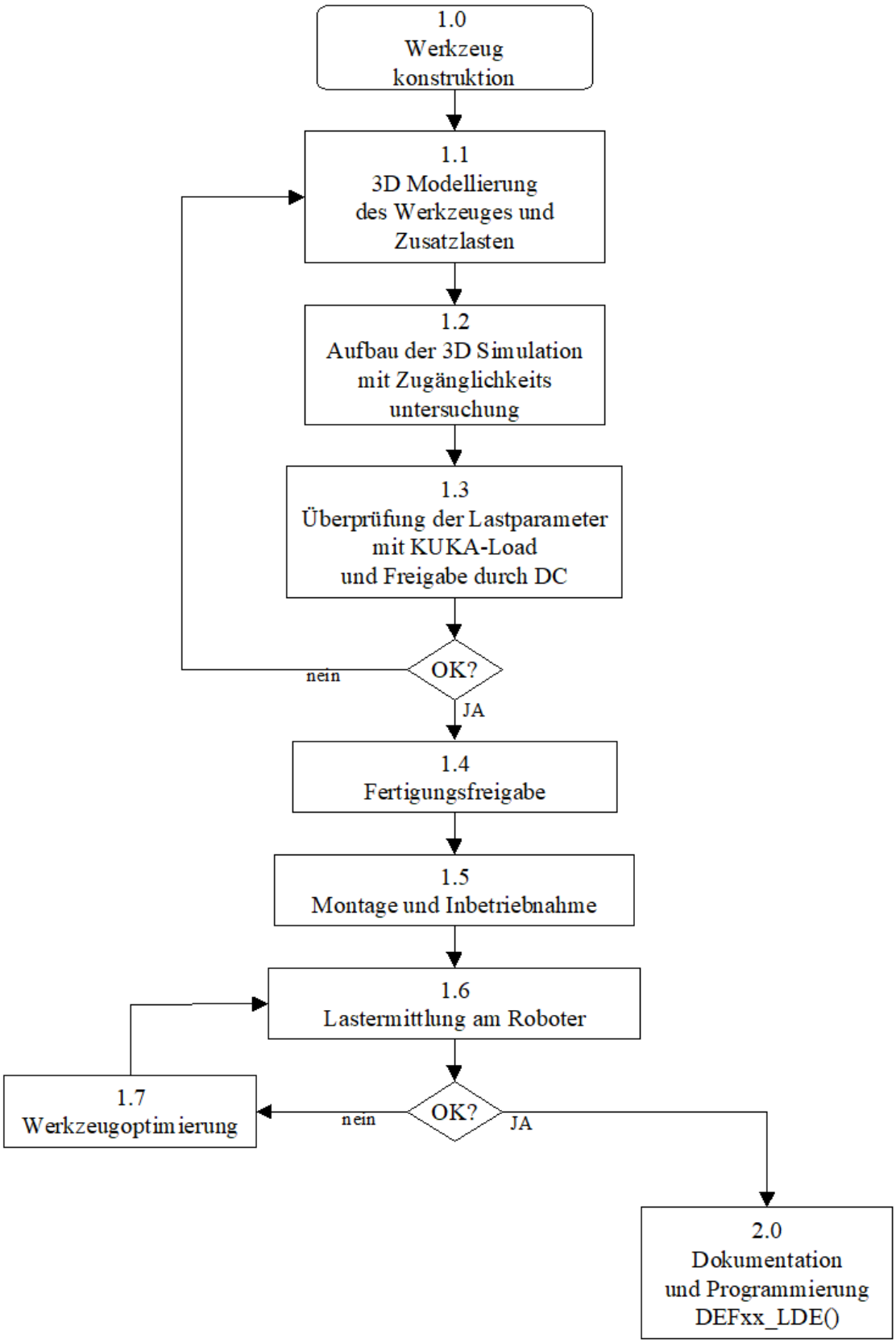

Figure 18 Load data determination procedure by contractor

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3 Utilities/service programs

3.1 General [V_70.00_I5]

Auxiliary programs are used to quickly analyze, start up and restore faults in the event of program offset. The utilities are located in the folder H_Prog. The programs described in the following chapters are included in the DAI_Baseproject as default programs and must be supplemented accordingly.

3.2 Pre-zero program [V_70.02]

Auxiliary program is executed without collision commands, as only the T1-Modus is run.

In the "dai_before_zero_mam.SRC" program, the mechanical zero and the pre-adjustment position are programmed with the tool in the system in order to be able to perform any necessary adjustment faster. When replacing the IR control or malfunctions (offset), the IR can be moved to the adjustment position using this program

The position of the pre-adjustment position (notches) is already preset as "PTP VORJUS" for the 6-axis of the robot from the machine data; only the additional axes must be stored in the "PTP VORJUS_E1" pre-adjustment position to" PTP VORJUS_Ex "in the" dai_before_zero_mam "in the" ". In the case of welding tongs, it is essential that these are docked at the start of the program in order to prevent this co-driving and damage.

The program must be created before the user programs and the conditions in the system taken into account. If not all axes can be moved to the pre-adjustment position at the same time, the corresponding axes must be moved by hand during adjustments.

If not all axes can be moved to the pre-adjustment position at the same time, the pre-zero must be programmed so that A2-A6 can be approached with the pre-zero position. In this case, this is to be represented with a comment in the program. For all other constellations, coordination with the Mercedes-Benz Robot Project Team is required. "Dai_before_zero_devided" file. All positions must be taught in here and cannot be accessed on the machine data!

The program must be created with the basic and tool correction data "Zero" (=BASE 64, TOOL 64, EXTERNAL_TCP FALSE) so that the program is independent of the tool.

3.3 Control measurement of robot mechanism [V_70.05]

The control measurement is used to check the geometry of the robot.

Within the auxiliary program "DAI_ROBOT_OK.SRC", the position of the inserted tip must be entered in the marked position so that it meets the fixed position. It must be clearly recognizable whether the geometry of the robot is OK.

The program also checks the connection of the hose package to axis 6in order to be able to restore it when changing the hose package.

For this purpose, the pipe is first moved to the extension in order to obtain the alignment and then the bore/grain size on the hose package holder, see marking in Figure 19.

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For this purpose, the example program is used as a specification for the "DAI_ROBOT_OK" as well as specifications for measuring tools.

Figure 19 Check “tube in tennis racket” and “hose package holder”

3.4 TCP ~~Tool~~ control program [V_70.06]

Auxiliary program is executed without collision commands, as the program can only run in T1-Modus.

The "dai_Check_ < Tool name. >. SRC" program checks the robot and tool geometry in the system by moving off the control tip. The program only applies to joining tools that have a TCP point. Gripper shall be tested as in GL_Zusatz (Volume 3 Chapter 40.01.01).

This control program must be created before component programming to be able to correctly diagnose a supposed program offset during programming phase.

If several (physical) tools exist, a separate program must be created for each of them.

DAI_check_ < Tool name. >. SRC. => Example: SWZ_01_C=> C-tongs

< Tool name >=> See Chapter 5.5.5 Tool names [V_70.00_I5].

~~Note: As long as the DAI_Checktool is not yet the 1 st. Control measurement with tool 64 can be used, the procedure must be adhered to as described here. In future, this will be created automatically via the DAI CheckTool program.~~

3.4.1 Robot-guided tool [V_70.02]

Auxiliary program is executed without collision commands, as only the T1-Modus is run.

The "DAI_Robot_Ok" UP "" is first called in the "dai_check_ < Tool name. >. SRC" program. In the first point, the control tip of the flange fixture on the robot shall match the system test tip. The length of the test probes (tool & system) used shall be entered as a comment in the test program. The same applies to the spacer rings, which are to be entered in the program with a comment of the diameter (mm). In the second point, the hose package holder is checked by A6 ("tennis racket"). These robot control points shall be saved with TOOL and BASE_DATA [64].

The first checkpoint in the "DAI_Check_ < Tool name >. Src" checks the tool geometry. It shall be programmed with TOOL/BASE 64.

The TOOL belonging to the tool must then be programmed with the same point! It is important that the first checkpoint with TOOL/BASE 64 must be programmed again afterwards

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To check the opening dimension of the welding gun, the counter electrode must be programmed with its own control point and the associated TOOL.

The DAI_Checktool is then called where the subprograms are automatically created for this purpose and the other checkpoints are taught. It must be remembered that the master reference must be created during initial calibration!

For positioning accuracy robots, the points taught with TOOL 64 are not positioning accuracy, as the load data are set to zero. However, this has no influence on the repeatability of these positions.

1. Insert joining tool into fixture without test ring and teach in position

   The upper edge of the tool must be retracted to the edge marked in the picture (red arrow).

2. Insert test ring-this must be possible without applying force, teach in if necessary
3. Tool control
4. Before extending the joining tool, remove the test ring

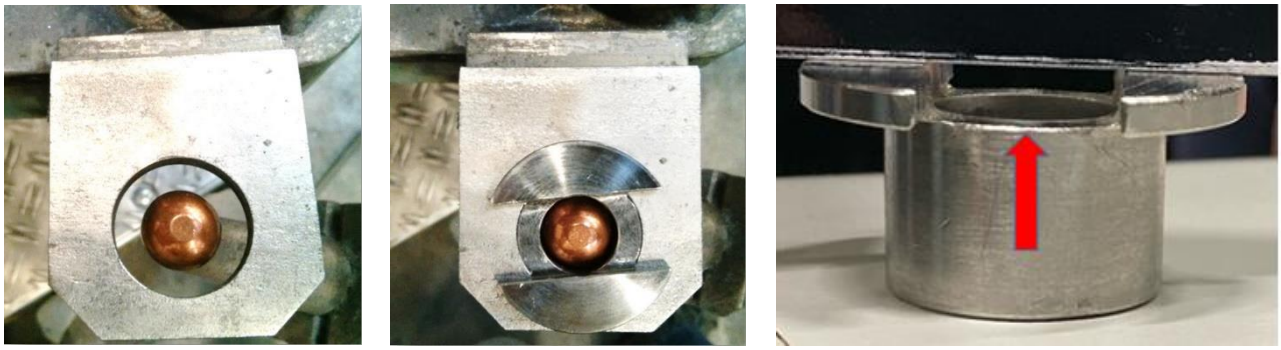

Figure 20 Procedure for test with test ring

3.4.2 Stationary tools [V_70.02]

Auxiliary program is executed without collision commands, as only the T1-Modus is run.

Check robot as 3.4.1 described in Chapter, then the external tool is checked by driving the control tip of the gripper onto the control tip of the external tool. This point shall be programmed with BASE/TOOL 64 and then with the BASE belonging to the external tool. A device for the control tip or the test ring can be mounted on the gripper.

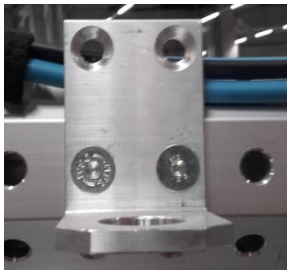

Figure 21 Flange mounting of external tools

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3.4.3 Tools for tool testing [V_70.05]

A stationary measuring tip shall be installed in plant-specific design for each robot. It shall be manufactured according to the drawing. It is used to check the tool work point (TCP) and the geometry of the robot.

The possible project variants can be found in the documents "Volume 2/Robot" in the project ECM.

These are specified for the specific project and can also be executed differently in the individual plants. This must be clarified in the design meeting.

This includes:

- ⇨ How many measuring tip sets/measuring stands are to be provided per system
- ⇨ How is storage planned

The tip shall be easily accessible and directly accessible for the components to be checked. The location must be selected in such a way that the production operation is not affected and there is no risk of injury to the employees.

Measuring tips shall be designed so that the shortest possible can always be used!

The installation location of the test probe shall already be taken into account in the system layout and coordinated with the client before assembly.

For stations where there is no fastening option for the above measuring tip due to the geometric circumstances, a measuring stand shall be used as an alternative. The definition of what is used for a variant shall be coordinated by the contractor with the Robot Project Team of the client.

When removing this measuring tip or the measuring head for the operating procedure, it shall be ensured that a fixed mount remains in the system for fixing the measuring tip. It shall also be ensured that all aids (tips and measuring stands) used in the model series are geometrically identical, so that the control programs can be run off with all existing aids.

These aids are to be labeled according to the designation system.

The various approved measuring tips can be found in the MFL.

See also GL-DE_Roboter_Supplement in Volume 3 Chapter 040.01

The necessary measuring tips for tool and basic measurement are part of the contractor's scope of delivery and shall be handed over to the client's Robot project team in a suitable container at the end of the project.

3.4.4 General Specifications [V_70.00_I5]

All control points used in the program must be documented. The exact designation of the checkpoints to be approached as well as the designation and geometry of the control tips used shall be marked with comment lines **in the program**.

After each checkpoint, the instruction is

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WAIT FOR $IN [1026]

To program to prevent further driving after the check point.

3.5 Gripper test program [V_70.00_I5]

See GL-DE_Roboter_Zusatz in Volume 3 Chapter 040.01

3.6 Sensor Tests

3.6.1 Sensor exchange [V_70.00_I6]

To be able to restore sensors that work with light, laser light or similar, a predefined position must be approached in the test program. These positions can represent crosshairs or similar alignment elements!

The first position must approach a defined value, which is also commented in the program (distance value in mm). This position serves as a replacement position for the sensor.

The second position is used to check whether the measurement value changes. The comment shows which measurement value (distance value in mm) is displayed and that the sensor was installed in parallel.

This shall be implemented as an auxiliary program!

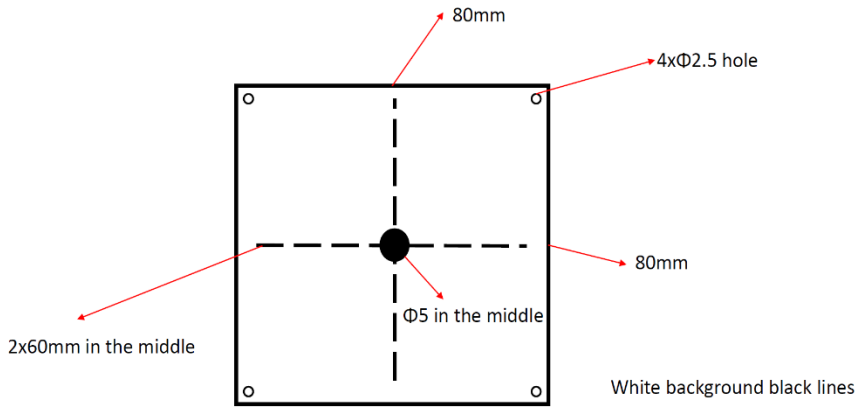

Figure 22 Example of crosshair sensor test [V_70.00_I6]

3.6.2 Sensor check [V_70.05]

In the case of sensors such as camera, measuring systems and similar, service main programs shall be provided that are then started cyclically from the PLC to calibrate themselves. Which main program should be used for the sensor check can be found in Chapter 4.4.2.1.

If not separately regulated in the GL documentation of the technology, the calibration plates shall be designed so that they are protected against dirt and crash.

Main program	Auxiliary program name	Assignment
HPxxx ()	Sensor_Check ()	Check program for sensors.

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Table 7 Assignment of Sensor_Check/English (standard)

3.7 Load data start program [V_70.00]

In the "dai_ldd_xx.SRC" auxiliary program, the travel distance from the HOME position to the start position of the load data determination shall be programmed. If the starting position cannot be approached directly from the starting position, corresponding intermediate points must be taught in.

If several different start positions or work areas are necessary, a separate program shall be created.

Furthermore, the selected work areas of all axes must be stored as a comment when determining the load parameters.

If no load data determination could be carried out on this robot, due to space or identical robots, this shall be commented on in the auxiliary program but no movements shown.

3.8 Maintenance program [V_70.00_I6]

The maintenance item should ensure the accessibility of the tools when changing the electrode cap and for other maintenance operations.

For maintenance work on the robot and on the tool, the movement sequence from the HOME position to the maintenance item shall be programmed in the "Maintenance0" program to" Maintenance6".

The maintenance program is called up by the PLC.

3.8.1 Tool change maintenance program [V_70.00_I6]

The program HPxx0 () is to be used for robots with fixed tool and robot with interchangeable couplings without tools (undocked). HPxx1 to HPxx6 shall be used for interchangeable tools 1-6. The comment on the programs must be clearly assigned. The traversing points, docking and undocking as well as the maintenance position shall be taught in the program MAINTENANCE0 for robots with firmly attached tool and robot with interchangeable couplings without tools. MAINTENANCE1 to MAINTENANCE6 is used for interchangeable tools. In the case of robots with docked tool in the basic position, the tool shall be docked at the start of the maintenance program and connected again at the end. The movements must be programmed so that the maintenance position can be reached without collision with each tool.

After reaching the maintenance position, the intended output must be set on the robot side (maintenance position reached). Subsequently, the IR is moved back to its basic position by a restart.

If the basic position (PTP Home) of the maintenance item corresponds to, an additional movement point must be inserted in the respective program Wartung0 to Wartung6 which moves the robot away from the home position, that the PLC locking signal "In_Home" is reset.

The maintenance items in the HPxx0 program are stored globally in "Global_points" as "XGMaintenance0" to" XGMaintenance9" so that this can be made available in advance for the OLP and VIBN. The associated FDAT is saved locally as with the home positions.

Main program number assignment can be read in Chapter 4.4.2.

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Main program	Utility Name (s)	Assignment
HPxx0 ()	Maintenance0 ()	Robot with fixed tool. Robot with changer without docked tool.
HPxx1 ()	Maintenance1 ()	Robot with docked interchangeable tool 1
HPxx2 ()	Maintenance2 ()	Robot with docked interchangeable tool 2
...	...	...

Table 8 Overview and assignment of maintenance programs English (standard)

3.8.2 Maintenance program functions [V_70.00_I6]

These programs can be used to start centrally from the PLC to execute function relevant for maintenance. No robot movement is necessary for this.

For example, all station covers can be opened here to clean them and then all closed after restart. Call via a separate main program.

The program HPxx0 () shall only be used if only one function is to be performed in which NO movement movement of the robot is carried out and the function can be set to basic state after restarting by the PLC.

The comment on the programs must be clearly assigned. The required travel points shall be taught in. The movements must be programmed so that the maintenance position can be reached without collision with each tool.

After reaching the maintenance position, the intended output must be set on the robot side (maintenance position reached). Subsequently, the robot is moved back to its basic position by a restart.

If the basic position (PTP Home) of the maintenance item corresponds to, an additional movement point must be inserted in the respective program Wartung0 to Wartung6 which moves the robot away from the home position, that the PLC locking signal "In_Home" is reset.

The maintenance item in program HPxx0 is designed globally as "GMaintenance" so that this can be made available in advance for the OLP and VIBN.

Main program number assignment can be read in Chapter 4.4.2.

Main program	Utility Name (s)	Assignment
HPxx0 ()	MaintenanceF0 ()	Maintenance function 0 (only 1 function and no robot procedure)
HPxx1 ()	MaintenanceF1 ()	Maintenance function 1
HPxx2 ()	MaintenanceF2 ()	Maintenance function 2
...	...	...

Table 9 Overview and assignment of maintenance programs English (standard)

3.9 Cleaning program [V_70.05]

The cleaning position should enable the cleaning of tools or, in the case of tools that may not be cleaned, bring them outside the cleaning team area.

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If no project-specific specifications apply, all robots shall be moved upwards in the cleaning position so that persons are no longer endangered!

For cleaning work on the robot and on the tool, the movement sequence from the HOME position to the cleaning position shall be programmed in the cleaning program. The cleaning position shall be created as a HOME [5] position!

The cleaning program is called by the PLC.

A tool change shall only be provided if a tool is arranged as specified in the specifications. This is e.g. If the tool must be moved out of the cleaning area, because highly sensitive sensors may not be cleaned.

The following procedure applies to this:

In the case of robots with docked tool in the basic position, the tool shall be replaced at the start of the cleaning program against the desired tool and the tool shall be connected to the basic position again at the end. The movements shall be programmed so that the cleaning position can be reached without collision with each tool.

After reaching the cleaning position, the robot automatically sets the parameterized output for Home [5]. Subsequently, the IR is moved back to its basic position by a restart.

If there are tools that may not be cleaned, this must be docked and brought out of reach of the cleaning. So that the area of the home position of a linear axis can also be cleaned, it must be moved away from the home position for cleaning.

The cleaning program is also used for power shutdown functions. It is therefore essential that the cleaning program sample is used!

If the basic position (PTP Home) of the cleaning position corresponds to the cleaning position, an additional movement point shall be inserted in the cleaning program which moves the robot away from the home position, that the PLC locking signal "In_Home" is reset.

Main program number assignment can be read in Chapter 4.4.2.

3.10 Tool change program for maintenance operations [V_70.00_I5]

The "dai_chg_ < Tool name. >. SRC" program serves as a tool for changing robot tools quickly. If a tool change is necessary, this program is selected in T1 mode and the robot from the HOME position to the tool change position e.g. Transport vehicle driven. Before the actual tool change position, an intermediate point must be programmed to then drive to the target point at reduced speed.

After the tool change has been completed, the system returns to HOME position.

The tool change point shall be selected such that the corresponding tool means of transport (e.g. Tool trolley) positioned easily in the system and the assembly and disassembly process can be carried out.

Whether the program is used must be coordinated accordingly in the design phase with the operators of production and maintenance.

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3.11 Caster measurement protective devices [V_70.00_I6]

Caster measurement on protective device with non-contact protective devices, the automatic programs are used.

See also Chapter 3.13 Safe Operation Programs [V_70.00_I6].

3.12 Check_Tool [V_70.00_I6]

For all tools "$Tool [x]", the contractor shall create these in Dai_Checktool, exception tools!

See description for DAI_Checktool.

3.13 Safe Operation Programs [V_70.00_I6]

See documentation Volume 3 Chapter 040.01.04 Safe Operation.

3.14 Brake test [V_70.00]

The automatic execution of the brake test requires that robot locks are planned to prevent crash.

The spline move command for the park position must be deleted and replaced with a new PTP.

The name of the parking position must remain the same!

3.15 Reference trajectory for predictive maintenance

3.15.1 Background [V_70.03]

Based on a coordinated and hard-defined robot movement (aka trajectory), anomalies and predictions with regard to mechanics should be derived in the medium term.

The possibility of a so-called reference trajectory - which should always be cyclically executed identically in the robots - is a possible basis for deriving and recognizing such anomalies and predictions.

Program should be run 1 time a day and is called by the PLC. However, it can change within the projects and over time. In the KSS environment 8.2 and 8.3, the frequency of execution must be discussed in the project (e.g. once a week) based on known runtime issues of these KSS versions.

The program "DAI_ReferenceTrajectory" is installed and managed by means of the tech package DAI_MQTTClient with the controller. The program must not be changed.

3.15.2 Process [V_70.03]

1. HP is selected by PLC (HP017 - see PRL in the respective project)
    a. Tool is disconnected if necessary
2. In HP, the application program "DAI_Trajektorie" is called and a start position is taught after the home position where the robot starts the movement

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3. In the module "DAI_Trajektorie" the call "DAI_ReferenceTrajectory( )" is executed. The UP must not be changed and performs the following topics.
    a. check if there is enough space and if a KL is available.
    b. starts the trace with the appropriate trace configuration
    c. each axle travels at different speeds and with/without stop
    d. Stop and send the trace
    e. returns to the starting position.
4. HP End

3.15.3 Hints [V_70.05]

1. Program must always be run with the same load (with Toolchange without tools)
    a. After uncoupling, if necessary, drive on Home if robot is not on Home after decoupling
2. A logical decoupling of a spot welding gun is not necessary
3. The program "DAI_Trajektorie" have to be stored in the service folder on the robot (R1/Service).
4. Collision detection :
    a. After Home → Dataset 30
    b. Before reference movement → Dataset 29
5. The following free spaces are required for each axle - starting from the starting position
    a. A1 to A3 → +- 5 degrees
    b. A4 to A6 → +- 10 degrees
    c. E1 (if KL) → +- 200mm
6. Duration of the entire program: ~ 1min

3.15.4 Implementation [V_70.06]

The example programs mentioned below serve as orientation during implementation on the robot.

- The main program and application program follow the normal programming conditions of Mercedes.
    - Possibly deviating project topics (e.g. collision detection) must be taken into account.
- The starting position should be as close as possible to the home position.
- Interlocks (not listed below in the examples) must be put on and used if necessary. Whenever a collision would be possible, interlocks (robots <-> robots or PLC <-> robots) MUST be used.
- In general, the movement must always be carried out collision-free.
- The last movement point (e.g. "PTP start") before calling DAI_ReferenceTrajectory() must match the appropriate tool for the current tool.

HP017( )	**DAI_Trajektorie( )**

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```
 6
 7  ; Anlagenbezeichn:   Stations-Nr:
 8  ; Stand / Version:
 9  ; Name Bearbeiter:
10
11  ; Nach Programmierung lÃ¶schen
12  ; delete after commissioning
13  wait for false
14
15  ; !!!
16  ; Kollisionserkennungen und Verriegelungen mÃ¼ssen en
17  ; Collision detection and locking must be carried out
18  ; !!!
19
20  DAI_HP_START
21
22  ; Home
23  PTP HOME Vel=100 % DEFAULT
24
25  ; Coll Dataset 30 after Home
26  COLLDETECT UseDataSet DataSet[30]
27
28  ;UP Aufruf
29  DAI_Trajektorie()
30
31
32  ; Coll Dataset 30 vor Home
33  COLLDETECT UseDataSet DataSet[30]
34  ; Home
35  PTP HOME Vel=100 % DEFAULT
36
37  COLLDETECT Off
38
39  DAI_HP_END
40
```

```
17
18  ; !!!
19  ; Kollisionserkennungen und Verriegelungen mÃ¼ssen en
20  ; Collision detection and locking must be carried out
21  ; !!!
22
23  PTP HOME Vel=100 % DEFAULT
24
25  ; Coll Dataset 30 Nach Home
26  COLLDETECT UseDataSet DataSet[30]
27
28  ; Only at tootlchange
29  ; Nur bei Werkzeugwechseler
30
31  ; 1/ --> Decouple tool
32  ; 2/ |-> Check for tool 0
33
34  ; auf Home fahren ggf. nach Decoupling wenn  Roboter
35  ; move to Home if robot is not at Home after decoupli
36
37  ; Coll Dataset 29 zu Start pos
38  COLLDETECT UseDataSet DataSet[29]
39
40  ;PTP/LIN zu start pos
41
42
43  ; Startposition
44  PTP start Vel=100 % PDAT1 Tool[1] Base[0]
45
46  ; Start Referez-Trajektorie
47  DAI_ReferenceTrajectory( )
48
49  ; Coll Dataset 30 vor Home
50  COLLDETECT UseDataSet DataSet[30]
51  PTP HOME Vel=100 % DEFAULT
```

Figure 23 Main- & Application program Reference-Trajectory

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4 Program structures

4.1 Project structure WorkVisual

4.1.1 Project name WorkVisual [V_70.02]

⇨ See also Chapter 1.9 Naming of robot system [V_70.02]

The name of the active project is composed as follows:

< System name > _ < Station > _ < Robot name >

Example: AS40_010RB_100

As of the status "Automatic mode", no more than 2 projects may be located on the robot, one of which is active.

4.1.2 Cell definition WorkVisual [V_70.02]

⇨ See also Chapter1.9 Naming of robot system [V_70.02]

The name of the cell is composed as follows:

< System name >

Example AS40

Figure 24 Cell definition WoV

4.1.3 Control name in WorkVisual [V_70.02]

⇨ See also Chapter1.9 Naming of robot system [V_70.02]

The name of the active project is composed as follows:

< System name > _ < Station > RB_ < Robot name >

Example: AS40_010RB_100

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Figure 25 Control name in WoV

4.2 General [V_70.03]

This chapter describes the program modules available in KRC5 and MBAG specifications as well as the underlying process concepts and naming conventions.

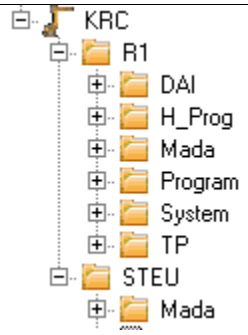

Figure 26 Folder structure in robot

Folder	Description
DAI	Basic functions packed in libraries (message handler, ghost, stop point end, etc.). Global data list for basic functions
H_Prog	Utility for manual operation (robot okay, tool-okay, etc.)
Mada	Machine data - no changes allowed
Program	All user and main programs
Service	Various service programs (cleaning, maintenance, ...)
System	Folder for system programs
TP	All technology packages (Mercedes-Benz & KUKA) and the associated data lists of technologies. Each technology is stored in a separate folder. Mercedes-Benz technology packages are located in a folder of the following names: DAI_ [Technology name] E.g. "DAI_HSN"

Table 10 Folder structure in robot

The user works with his programs exclusively in the directories:

- Program (programs for automatic operation)
- H_Prog (programs for manual operation)

Changes may only be made to a limited extent in the KUKA-relevant directories /R1/System and /STEU/MADA. The conditions for this are:

⇨ Software limit switch via menu
⇨ Entry of $Tool and $Base data, as well as associated load data

Directories may not be created outside the folder "Program" and "H_Prog"!

Programs installed by a tech package or base system may not be moved or changed.

4.2.1 Import of data into user files [V_70.03]

If in ready-made user files e.g. "Global_Points.dat" changes are to be made, it is mandatory to use the function of "merging" data from two sources in the WorkVisual.

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This ensures that important jump marks are not destroyed by the Baseproject for updates!

See KUKA's WorkVisual documentation (documentation can be accessed via Xpert.kuka.com).

4.2.2 Dai folder [V_70.03]

Do not make any changes in the /R1/DAI folder if the file begins with "DAI". In the following "user files", however, entries may be made. These files will not be overwritten even with update!

Folder	File	Hint	Function
/R1/DAI	user_settings.dat	Settings	Settings that the user may/must make
/R1/DAI	user_sps_ini.src	PLC. Sub-calls (init)	Call for deselected program to the default position
/R1/DAI	user_global.dat	Data storage for entries user	Storage of global data (formerly in $config.dat)
/R1/System	Global_points.dat	Data storage for entries user Position data	Storage of global data For Global Motion Points

Table 11 contents of the user-files

4.2.3 H_Prog folder [V_70.03]

In the folder "H_Prog" all subroutines must be set, which are ONLY run in T1/manual mode, e.g. dai_Tool_01_CHK.src. Important, otherwise this will be recognized as an error in the WoV check routine!

Do not store subroutines for automatic operation in the "H_Prog" folder!

Collision detection does not need to be programmed for utilities.

4.2.4 Program folder [V_70.03]

The Program folder stores all major and global subroutines used for the automatic run. In this folder, subroutines with the same functions can be combined into subfolders for better clarity (e.g. many gripper programs).

Here is an example:

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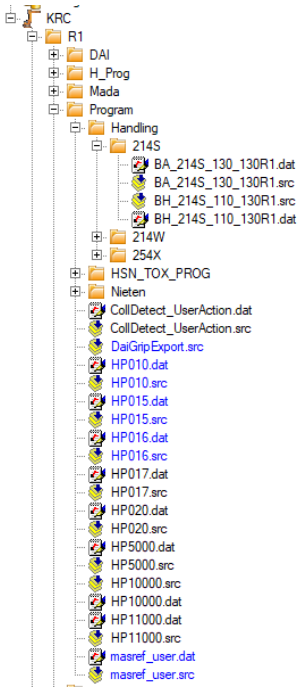

Figure 27 Example sub-folder structure in the "Program" folder

4.2.5 Service folder [V_70.03]

The "Service" folder stores programs that are managed by the base project or other global functions.

These must not be postponed under any circumstances!

These include, for example:

- Cleaning

4.3 Program flow [V_70.05]

In automatic external operation, the user programs are called up and started by the system control (PLC) via the program number (HPxxx). In the KRC, the programs P00_SUBM.src/dat (program number transfer) and SPS.SUB (program selection) are required for this.

The SPS.SUB program runs as an independent process in the background and handles the program number transfer from the PLC.

If the transmission is successful, the associated main program (HPxxx) is started, which in turn calls up the corresponding global subprogram.

Example of HP2010.SRC on robot 050RB_200:

DEF HP2010 ()

I I

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; Equipment identification:

; Change status: 01.12.2020

; 12/1/2020 8:00 NAME: SMITH CHANGES:

PTP HOME Vel=100% DEFAULT

COLLDETECT UseDataSet DataSet[30]

Conditional stop 16

BH_050R2_040_W_RL (); Fetch from station 40 variant W right-hand drive vehicle

Conditional stop 16

SWB_ 050R2_050_W_RL (); stud welding in station 50, variant W right-hand drive vehicle

Conditional stop 16

BA_ 050R2_060_W_RL (); storage in station 60, variant W right-hand drive vehicle

COLLDETECT UseDataSet DataSet[30]

PTP HOME Vel=100% DEFAULT

END

=> Designation shall be executed in accordance with national language! See following pages

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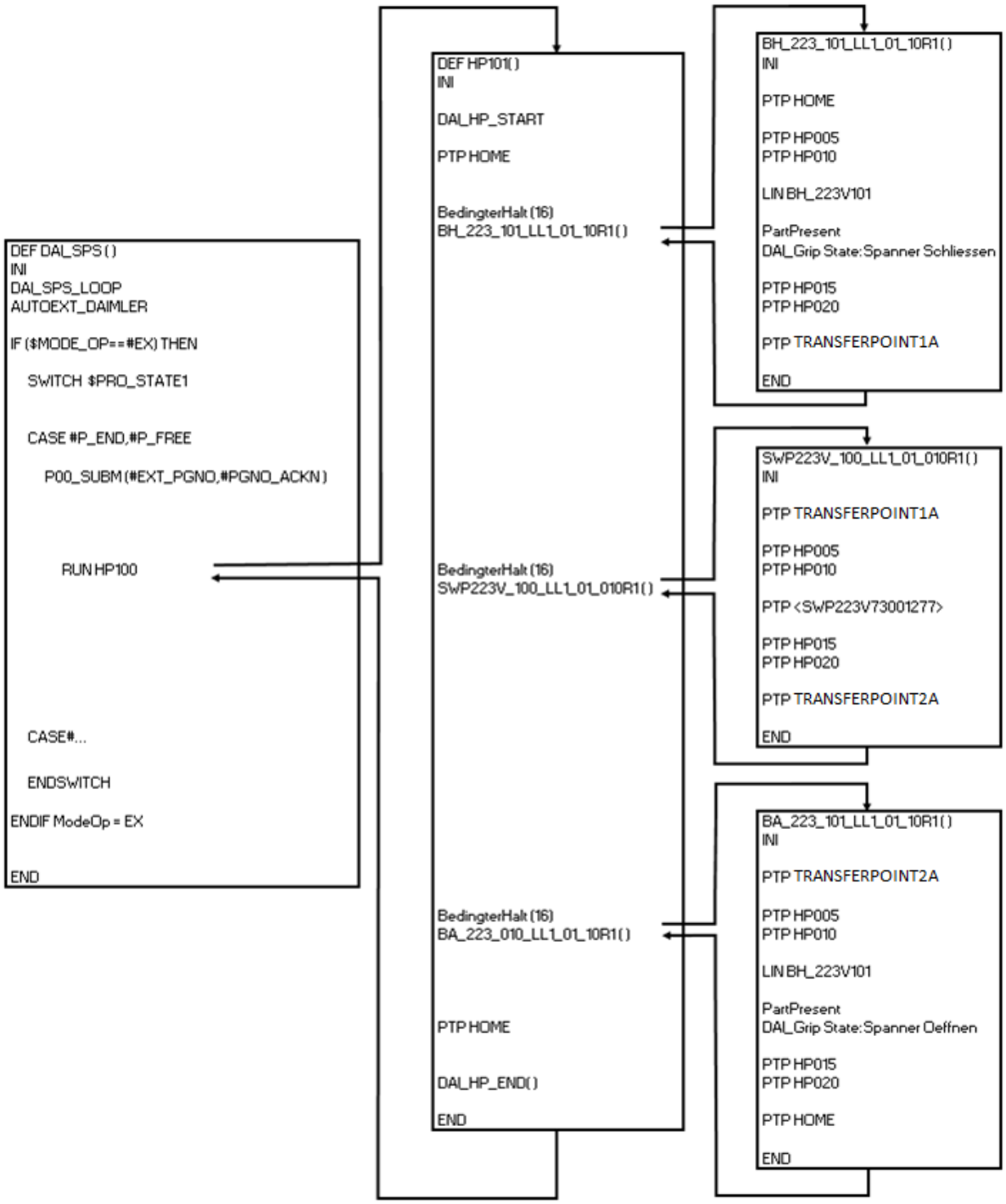

Figure 28 Program flowchart

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4.4 User programs

4.4.1 General [V_70.00_I5]

User programs are programs programmed by the contractor. These include main programs, global subprograms and special programs.

All commands shall be programmed via inline forms.

4.4.2 Main programmes (HP) [V_70.05]

In addition to programming initializations and branching logic, only global subprograms are called in the main program. It is not permissible to program movement commands except the HOME position.

in area HP001 to HP099 (e.g. Maintenance item, channeling in/out,...).

When creating a new main program, make sure that the template VG_HP (softkey) is selected. The exception is the HP cleaning that must be implemented as per example!

The name of the program must be entered in plain text as a comment (maximum of 30 characters). Ensure that only valid program names (HP001.SRC-HP32767.SRC) are used.
In the event that a iw7-Roboter is installed in ReUse systems of iw4 or iw5 the number of main programs shall be reduced due to the interface width.

The assignment of HP to component programs shall be specified by the contractor before the start of the project with the Mercedes-Benz project team Control Engineering (not robot) of the respective plant and are documented in Volume 2.

The designation "HP" is implemented in all plants irrespective of the national language!

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4.4.2.1 Main programs/special standard programs [V_70.06]

For Integra 7th phase, the main programs are reallocated.

From HP1000 project-specific=> see previous chapter.

Main program	Assignment
010	Cleaning program
015	Brake test
016	Adjustment reference run for Safety SOP~~/SRM~~
017	Reference trajectory for predictive maintenance
020	Maintenance program without tool changer or tool coupling without docked tool
021-029	Tool maintenance program (1-7) + placeholder
030 – 039	Special function maintenance program e.g. Flaps stations on/closed
040 – 041	Spot welding specific programs (cap change)
040 – 041	Laser-specific programs (PCU/FCU)
040 – 045	Arc welding-specific programs
041 – 045	FlowDrill screws Weber (SRF) service programs
051 - 052	Stud weld specific (lamella check)
060 – 069	Placeholder for technology-specific
070 - 079	Running empty/driving
080	Sensor check
081 - 089	Placeholder for project-specific special case
090	Adhesive rinsing program assembly (see GL add-on assembly, assembly line 3, chapter 40,440) or bodyshell "collective rinsing"

Table 12 Assignments of main programs/standard standard programs

4.4.3 Global subprograms [V_70.00_I5]

In global subprograms, the actual traversing points and process commands are programmed depending on the task. You are never called directly in external operation, but always via the corresponding main program (HP).

Depending on the technology used (s) or task, the programmer must assign the program names of the following convention:

4.4.3.1 Simplification of programs at equal process points [V_70.00_I6]

In the future, joining points are differentiated from the vehicle/component into derivative points. In the robot, these joining points are always provided globally by the simulation and must remain globally.

4.4.3.2 Naming Conventions of Global Subprograms [V_70.00_I6]

The program names of global subprograms shall be assigned as follows:

Example: SWP206_020_*Option*_01_010R3

SWP	206	_020	_Option	_01	_010R3

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Technology abbreviation (max. 3 Characters)	Optional: Type e.g. 206, MRA2, etc. (Max. 4 Characters)	Optional: Station number (max. 3 Characters)	Optional: Load stages, design, etc. (Max. 7 Characters)	Optional: Consecutive counting	Robot number (shortened by 010RB_300)

Table 13 Naming conventions for global subprograms

⇨ If more than 24 characters are used, this must be coordinated!

To execute all names & comments in national language accordingly!

Example: SWPMRA2_120_01_010R3

Subprogram code for action with component:

Abbreviation	Technology
BH_	Fetch component
BA_	Put down component
BF_	Join component (mainly assembly)
...	Etc.

Table 14 Subprogram code action with components in German

English designations see English document.

Technology abbreviations shall be implemented according to defined designation system, Volume 3, Chapter 04!

Example of technology abbreviation:

Abbreviation	Technology
SWP_	Spot welding
SWB_	Stud welding
KLB_	Bonding/seam sealing
...	Etc.

Table 15 Example technology abbreviation

Technology abbreviations shall be implemented according to defined designation system, Volume 3, Chapter 04!

If other technologies are used, this shall be coordinated with the MBAG project team.

Consecutive counting: Continuous counting of subprograms of each technology. Each technology is counted separately.

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Station number: Refers to the number of the station where the program is executed.

Robot number: First position of the robot number

Options:

Here, vehicle variants (type) or other differentiating features such as. Versions are stored where a distinction is deemed useful in the name.

Definition of how subprogram structure is to be handled with global points in the project is to be coordinated project-specifically.

4.4.3.3 Program contents [V_70.00_I5]

Each program consists of 3 parts in the order:

- INI part (technology-specific agreement part) must not be changed. For offline programming, refer to template files 5.3
- Program header (must be filled in completely and completely. Equipment identification: Change status: 9.05.988:00 NAME: SMITH CHANGES:

=> Designation shall be executed in accordance with national language! See following pages

- **User initializations**
- **Instruction part**

The contractor shall change the information in the program header for his application. Unauthorized extensions are not permissible. The contractor can additionally insert its own agreements and queries in this part that are valid for all programs. After this, you can copy the changed program or program blocks into the other user programs and insert its program records between the HOME positions.

4.5 Technology Packages [V_70.00_I6]

MBAG provides programs for most technologies which are mandatory to use. They are loaded in the control system for use accordingly and must be parameterized by the contractor if necessary.

Technology packages update shall be implemented promptly by the system supplier as described in Volume 2. Urgent updates are communicated via the robot regular communication or project.

The following programs are available: (List is not complete)

KUKA-Specific Technology Packages:

- MAG welding (KUKA ARC_TECH) + Mercedes-Benz extension
- Laser welding (Kuka Laser Tech/LaserCut) + Mercedes-Benz extension

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Mercedes-Benz-Own technology packages:

- Handling (DAI_GRIP)
- Spot welding of servo gun from Bosch incl. Cap milling ,-change
- Stud welding
- Clinching
- Adhesive bonding
- Etc.
-

These technology packages were created with a certain firm, or software version of the external control. A function guarantee for subsequent versions cannot therefore be accepted. If an adaptation of the technology packages is necessary due to modified technology control systems, the contractor shall commission the Mercedes-Benz Robot Project Team with the change. The contractor shall bear the costs for the adaptation.

It shall be possible to program the technology programs from the user level via inline forms.

The functional descriptions of these packages are contained in the attachments of this guideline.

The SRC files of the technology packages must not be changed. Only the planned parameterizations in the DAT files may be adapted.
For acceptance, the versions of the technology packages approved by MBAG (only SRC files) are imported!

4.5.1 Technology Gripper [V_70.00_I6]

In global subprograms, the actual process points and process commands are programmed depending on the task. The gripper commands are described in GL "000_B3.040.01.01_GL-DE_Roboter_Zusatz_xx ".

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5 Programming

5.1 General [V_70.00_I5]

This chapter describes the conventions specified by MBAG regarding programming of the robot system. The basis is the software release 8. X.x of KRC5.

If newer software versions are available until the start of commissioning, these shall be used.

5.2 Energy saving functions

5.2.1 Energy saving driving style [V_70.00_I5]

The robot programming must ensure an energy-saving driving style.

This can be achieved, for example, by reducing accelerations/speeds and avoiding unnecessary movements.

5.2.2 Energy function "Set_Brake" [V_70.05]

Since the basic project 2.4.3.1 there is the possibility to have the brakes come into contact with the main program end earlier in order to save energy through a new inline form.

Currently, the inline form for the main program templates is already integrated in the function "DAI_HP_FUNC/HP-END" with the program end.

The ILF provides two parameters:

/ / Time When the robot is stationary, the brakes have occurred after this time has elapsed

Value range: 500ms to 20000ms (default & recommendation 2000ms)

/ / WithExAx7 It is decided here whether the 7. Axis (for linear units) should also be addressed.

Value range: YES/NO (defaukt: NO/With KL conversion possible to YES)

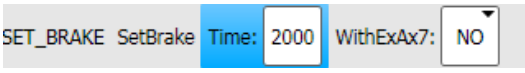

Figure 29 Inline form "Set_Brake"

5.2.3 Power shutdown [V_70.00_I6]

For power shutdown, the "Profienergy" function in the Profinet driver is implemented in the robot. The function can only be performed via the PLC on the robot without main program selection in the Homex position or in the cleaning program in Home5!

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5.3 Program creation

5.3.1 Create programs [V_70.00]

The new creation of a program can currently be selected between several alternatives. Make sure that the correct template is used! The template to be used are listed in the table below.

Template name	File name	Meaning
Main program	Vg_hp	Only for main programs
Application	Vg_appl	For global subprograms
Expert function	Vg_function	Subprograms with return value
Expert module	Vg_modul	Subprograms
Others can follow		

Table 16 Overview of existing templates

The name may be a maximum of 24 characters long and is to be assigned according to the conventions 4.4.3.2 of Global Subprograms [V_70.00_I6] described in Chapter.

Regardless of the access level, a maximum of 30 characters long comment that describes the component program must be inserted when creating a new program.

This comment is always visible in the navigator next to the program name.

Variable declaration is only permitted in the local data list (FOLD USER_DECLARATIONS). The fold must be created manually immediately after the "EXTERNAL DECLARATIONS" fold.

At the start and end of the program, either home points or transfer points must be used so that the transfer to the next subprogram is ensured at any time. This is to ensure that when the first movement point is canceled in a subprogram, it can be approached again!

5.3.2 Command Use in Programs [70.04]

- Interrupts, CYCFLAGS, flags and timers may only be programmed with the approval of the Mercedes-Benz Robot Project Team.
- User interrupt/cycleflag/timer in the area 101-105in coordination with the Robotics project team (Note: Number of active triggers instructions are limited (see KUKA documentation)!)
- Allocations to $Advance are not permitted.
- GOTO instructions are prohibited.
- EXOR queries are not permitted; the MBAG function "Select_one_out_N" is to be used for this. See GL Baseproject
- Movement records (except HOME) are not permitted in HP.
- In main and subprograms, the instructions may not be hidden in folds.
- No local subprograms may be used in the main program "HP ".
- No changes or assignments to base or tool (except for robot guidance)
- In programs the command "Continue" as well as the "Cont" function in logic inline forms, to avoid lead stops, are not allowed. Outputs can be programmed with a “synout”.
- Use of "zero frame" for movement points for tool or base is not permitted, for this purpose the base [64] and/or tool [64] must always be used.
- If no long texts are displayed in the logic command for inputs or outputs, comments must be inserted for labeling the inputs or outputs above the logic command (e.g. IF-loop).

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5.3.3 Gripper application [V_70.00]

The grippers are controlled and the components are queried using the MB tech packages provided.

In order to avoid collisions, the gripper actuator must be checked for the correct position before picking up a component. After gripping the component, the component query must be checked to ensure that the component was picked up correctly.

5.3.4 General data use [V_70.05]

Custom variables can be declared at the end of the "/R1/DAI user_global.dat" file. You can create your own sub-folds.

Entries in $config.dat are only permissible after coordination with the robot project team.

No changes shall be made at the Windows level. In particular, no own programs may be installed, system or network settings can be changed.

If data must temporarily be temporarily stored, this may only be done on the D drive: In the Commissioning folder. The data scope must not exceed 100MB. The contractor is always responsible for securing this data!

When the project is completed, the contractor must clean up the data list and remove all unnecessary points!

5.3.5 Program Mirroring Function [V_70.00_I5]

Should drain programs with space points be used for a system mirroring, the KUKA function "Axle Mirrors" may NOT be used for this purpose.

Offline programs that master these geometric offsetting are to be used for this purpose.

Mercedes-Benz specifications do not exist; it must only be ensured that the specifications of the program structure are complied with by the robot manufacturer (warranty in the event of misconduct is at the expense of the system builder).

5.4 General Programming Specifications [V_70.00_I6]

See also GL from the base project

5.4.1 Override [V_70.02]

The override must not be set automatically to a value in a robot program. If the application requires a reduced travel speed, it shall be programmed in the movement record.

Example:
PTP P7 CONT Vel=80% PDAT3

If the application absolutely requires a speed in the program override of 100%, the program override must be checked by calling the "CheckOverride" subprogram.

PDat modules are defined for assembly applications:

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See additional GL Volume 3 Chapter 040.404.01.01_GL-DE_Roboter-Erw.Assembly

5.4.2 Conditional stop [V_70.02]

64 conditional stops are available for programming a conditional stop.

BedStop [1] - stop [64]

A conditional stop is always to be programmed before distinctive positions and before each call of global subprogram in the robot program.

A prominent position is:

- Joining position
- Pick-up position
- Placement position

For querying the conditional stop, the subprogram "„BedingterHalt(x)“" shall be inserted at the corresponding points.

For a better overview, the conditional stops („BedingterHalt“) should be programmed under the "Variable overview" display.

BedHalt[1]= pick up position

BedHalt[2]= put down position

BedHalt[3]=Measurement (all measurement types, whether analog sensor or camera etc.)

BedHalt[4]=before each adhesive seam: Between the inline command "Adhesive Init" and

"Adhesive-fault stop" is technically not possible!

BedHalt[5]=before each weld spot & seam (no matter which procedure laser, MAG etc.)

BedHalt[6]=before each milling position

BedHalt[7]=before each stud welding

BedHalt[10]=all other joining options of technologies not listed

BedHalt[16]=before each subprogram call

The commissioning engineer can use BedHalt[11-15] for other prominent positions.

Much is also covered by “HaltPunktEnde“.

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For the BestFit application, the 32 conditional stop function is completely occupied and slightly different. (detailed description, see guideline DAI_BusFit)

Texts can be adapted in WoV in file “\\C\KRC\Data\DAI_USER_MSG.kxr”.

5.4.3 Ghost mode [V_70.00_I5]

A ghost operation shall be implemented to simulate the system sequence without components. MBAG technology packages are prepared for this purpose. Component queries and locks must be programmed accordingly. Details can be found in the documentation from the base project.

5.4.4 Function selection [V_70.02]

For applications in assembly and paint shop it is essential that a deselection of medium order e.g. Adhesive, NAD, SAM, etc. Is enabled via the PLC.

See additional GL Volume 3 Chapter 040.404.01.01_GL-DE_Roboter-Erw.Assembly

See additional GL Volume 3 Chapter 040.304.01.01_GL-DE_Roboter-Erw.Oberfläche

5.4.5 "Select_one_out_N" auxiliary program [V_70.00]

The "Select_one_out_N" function was developed because the Exclusive Or (Antivalenz) in robot programs was often incorrectly used.

See Baseproject documentation.

5.4.6 "Select_Distribution (Switch)" utility

The “Select_Distribution” function is built on the basis of the “Select_one_out_N” program. The program was developed to make programming switch instructions easier.

See Baseproject documentation.

5.4.7 "Select_Type" auxiliary program [V_70.00]

The “Select_type” function has been developed to program a switch case statement of the different Variant ID data types transmitted by the PLC. For example, based on the load level variants, a decision can be made as to which joining points should be made. This reduces the number of application programs required.

See Baseproject documentation.

5.4.8 Collision detection

5.4.8.1 KUKA collision detection [V_70.05]

The collision detection of KUKA is used in all movements in the robot programs.

See documentation:

- Manufacturer documentation

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- 000_B3.040.01.09_GL_EN-Collision_Detection_Vxx.xx

See additional information in Volume 3 Chapter 040.01.09 Mercedes-Benz General

Collision detection must be activated immediately during commissioning in order to protect the robot's tools right from the start!

The teaching takes place after coordination with the robot project team. All robots must be trained and run smoothly by the end of the “KFU”.

All movement instructions with gripper tools that take place in the immediate vicinity of other components, tools, storage/retrieval points, LCA magazines, SLT and transfer points to/from the conveying system etc. Shall be monitored by means of collision detection.

The collision detection shall be programmed so that the robot stops as soon as a greater force than in the normal procedure acts on the gripper. This prevents the gripper from being damaged or at least slightly damaged in the event of a collision.

The programming specifications can be found in guideline “000_B3.040.01.09_GL_EN-Collision_Detection_V70.xx”.

If the possibilities of the technology package are insufficient or unsensitive, a separate system must be installed for this purpose. This shall be coordinated beforehand with the Mercedes-Benz Robotics Project Team.

5.4.9 Locks [V_70.05]

The various types of system locking as well as collision interlocks between robot and robot are described in GL “Robot-PLC-interface-and-Baseproject” inside the base project tech package.

If “$Workspace” is used for locking, it should already be created in the OLP simulation.

The use of $Workspace must be clarified with the robot project team before the project starts!

Interlocks must be planned for wherever collisions with equipment or other robots can occur. This is to be implemented in all automatic and special programs.

If robots operate across stations, robot-robot locks 1-4 with monitoring (interrupt) must be used.

The definition of the locking signals can be found in the PLC sample program. I/Os with the same number must be used for the interlocks.

example:

release drive in process 1 100GSTxxx (IO121)

release drive out process 1 100GSTxxx (IO121)

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5.4.10 Acceleration/speed values for track movements [V_70.05]

For the technologies adhesive bonding, roller hemming, arc welding and laser welding, it is necessary to change the standard values for rail movements (LIN, CIRC) of the robot. With other technologies these may NOT be changed.

For this purpose, boundary conditions are given by Mercedes-Benz AG. The contractor shall not change these values independently, but only according to Mercedes-Benz specifications.

The parameters must be set by the contractor for each program using a separate inline form. The specification of the values to be set is determined by the simulation. The maximum values of the parameters should not be used across the board. During acceptance, Mercedes-Benz will check whether the values are plausible.

Functionality: Switch on program Start, switch off at program end.

Note: If the individual motion points are placed very close together, the speed is reduced according to the robot's rail planner.

Despite increasing the values, the robot may still not reach the set values. The cause may be due to poor spot distribution or excessive loads.

5.4.10.1 Non-Power Technologies (without Process Force) [V_70.00_I5]

In these technologies, no external forces shall act on the robot mechanism during application. This includes e.g. All adhesive processes.

Maximum value ACC_CP=5.00 and ACC_ORI1=500/ACC_ORI2=500

5.4.10.2 Force-acting technologies (with process force) [V_70.03]

For the technologies, external forces are generated during the application, which can only be calculated to a limited extent and thus act on the robot mechanism. In case of doubt, these can lead to destruction or damage to the mechanical components and thus to a loss of the robot manufacturer's warranty. This includes all roller hemming processes, for example.

Maximum value ACC_CP=5.00 and ACC_ORI1=500/ACC_ORI2=500

To achieve this for the operators of the plants in Mercedes-Benz AG and for the validation of the plant builders, the following boundary conditions shall be complied with:

- Only for process approved robot types may be used
- Provisions of the Process GL-DE e.g. Roller hemming with regard to programming specifications must be complied with
- In the simulation release, the responsible Mercedes-Benz project representative for roller hemming and robot technology shall be involved as to whether the set acceleration and speed values are technically feasible.

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5.4.10.3 Call up inline form [V_70.00_I5]

Call as inline form

Figure 30 "Path and orientation speed" call

Setting the values at the start of the program:

<u>Value range when setting:</u>

ACC_CP: 2.3 to 5 (default 2.3 as standard KUKA)

ACC_ORI1: 100 to 500 (default 100 as standard KUKA)

ACC_ORI2: 100 to 500 (default 100 as standard KUKA)

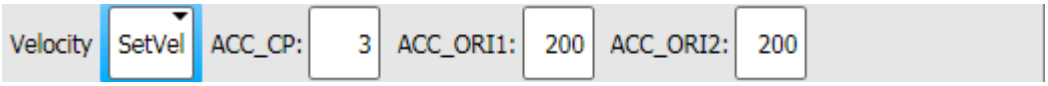

Figure 31 Inline form for setting "SetVel"

Setting the default values at the end of the program:

Figure 32 Inline form for resetting "SetVel"

5.5 Movement programming

5.5.1 Framework condition from simulation [V_70.00_I6]

The following boundary conditions shall be observed during commissioning:

(Takeover as extract from Volume 3.041.01.01 GL-DE Robot Simulation)

- Observe expansion joints

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- Observe robot hose package
- Provide min. 100mm accessibility reserve in all directions (ball with 100mm radius around each point)
- Observe singularity (check with RRS module)
- Overall procedure starts and ends at same space point with same axle configuration
- Observe cable ducts and other floor installations (or 350mm high floor component as interference edge for shop installation on floor, ground clearance OKF to ducts 200mm)
- 5% axle reserves must be complied with

Deviation from parameters is ONLY possible with approval of project team Simulation-Robot and system operator.

5.5.2 Boundary condition-FDAT for assignment of tools [V_70.02]

Within the Programm.src the tool and base assignment is always stored locally in the FDAT.

If a teach point is used multiple times within the program, it always has the same FDAT.

If a tool is changed and other load data is to be used in the background, a new point must be created for this.

Example at the beginning "Home" or "Home1".

5.5.3 Point names [V_70.00_I6]

Movement points are divided into classes depending on the scope and the process:

- Auxiliary points
- Process points
- HOME points
- Special points (e.g. PRE-ADJUSTMENT)

These points can be executed both locally and globally. In addition, there is a differentiation between Cartesian and axis-specific points in the control system. Cartesian points are stored in X, Y, Z, A, B, C, with reference to the $World - $Robroot - $Base - $Tool coordinate system chain. Axis-specific points, however, store the position of each axis in degrees or mm, depending on whether the axis executes rotational or translational movements.

In order to be able to easily distinguish the point classes in the offline programming or when uploading robot programs to a graphical offline programming system, the naming conventions of point classes as follows are defined as follows:

Example: PTP YYZZZ

PTP	YY	ZZZ
Motion instruction (not part of the name)	Point class	Number ZZZ

Table 17 Point names

5.5.3.1 Scope [V_70.05]

Point variables can be saved locally or globally. Local variables are only valid within the subprogram in which they were declared or newly created by the corresponding BOF command.

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Global points are valid in all program modules, and are used if the process points are required across all programs.

The coordinates of the global points are saved in "Global_Points.dat ".

Parameters such as PDAT and FDAT are also stored locally in the DAT of the associated SRCs at global points. This allows the request, but carries a certain risk (crash) in handling with status and turn. For this, collateral/mechanisms are included in the simulation and in the cooling lubricant, which can lead to damage in the event of disregard.

Information on handling global points can be found in the Kuka documentation.

Exceptions are the points "HOME" and "Transfer points" which are specified by the system.

The programmer is responsible for the correct scope of the points.

programmer is responsible for the correct scope of the points.

All points that were not defined as global points in this guideline are to be taught as local points.

All joining points linked to the vehicle/component data record shall be defined as global points.

Serial numbers are always in steps of 5in the OLP for projecting. After installing the OLP programming and optimization with the first components, the 5-step steps must be observed!

On the construction site up to the submission of BGB, the system manufacturer still has to display a minimum reserve of 2 points in between per 5-step.

Example: PTP HP005

PTP	HP	005
Motion instruction (not part of the name)	Point class here HP for auxiliary point	Consecutive number here 5

Table 18 Point names Scope

Example of local point: PTP HP010

Example of global point: PTP < HP010 >

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5.5.3.2 Point class auxiliary points [V_70.00_I6]

Auxiliary points are required to move from a specified position to the target position. They are necessary for moving obstacles or interfering edges in the system.

You will be marked by the character string HP.

Example: PTP HP005

PTP	HP	005
Motion instruction (not part of the name)	Point class here HP for auxiliary point	Consecutive number here 5

Table 19 Point class auxiliary points

5.5.3.3 Point class process points (example) [V_70.05]

Serves only as a sample. Description per defined technology is in the additional document of GL (000_B3.040.01.01_GL-EN_Robot_Supplement_Vxx.xx.pdf)

The names of process points include information about the scope of validity, the model series, the variant of the vehicle, the system number, the design group and finally the point number. You must build up the following schema:

It starts with point class, which is dependent on the application and can be found in the technology abbreviation description.

The vehicle type and then the model series type, i.e. "W", "V", "C", or "S" are then entered with "205" to differentiate between additional variants such as. RL=right-hand drive vehicles serve the next positions.

If no distinction is necessary, this name component can also be omitted. (See fields with option)

The system number (single-digit) serves as a differentiation criterion if this point is needed on more than one system.

The design group identifies the component or a component group and is given two digits. The point number, the number of digits depends on the technology and must be clarified with Mercedes-Benz before programming begins. These and the design group number can be taken from the point plan.

Process points/lanes can be implemented with the global points of the KSS function. These are then created via the OLP simulation and transferred to the robot.

If two independent extended joining point numbers are used in the project, the model series/project must be stored in the option.

To ensure that the global joining points can be treated unambiguously,

Only add the option information of the project to the process point.

Example of global process point: PTP < SWPMMA_62123456 >; PTP < FG_MMA_02123456 >

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Example of global process points if this has to be different due to different geowers and thus the difference in type: PTP < FG_MMA_W_0212345 >

Example: PTP FG_Opt_02123456_005

PTP	FG_	OPT_	02	123456	_005
Motion instruction (not part of the name)	Technology abbreviation here FG_	Option for additional info	Design group here 02	Joining point number here 123456	Optional: Consecutive number for e.g. seam applications Steps of 5 _S for Start _E for End

Table 20 Point class process points

Option can be:

Project e.g. "MMA", "EVA2"; Vehicle model e.g. „206“; Vehicle variant e.g. "W";

Steering variant e.g. "RHD"; System number e.g. 1; Etc.

Technology abbreviations shall be implemented according to defined designation system, Volume 3, Chapter 04!

5.5.3.4 HOME points [V_70.02]

HOME positions are created by the system for axis-specific (E6axis) and must not be. "E6pos "!

It is essential that each main program starts and ends with a HOME point.

It shall be ensured that collision-free driving can be carried out between all home positions. If this is not possible, application-related bypass programs shall be created.

6 HOME positions are available.

The exception here is the HOME5, which is reserved as a point for the cleaning position!

In the case of a tool changer, the check routine for tool changers must be called up before the first HOME point in the main program.

Note: When selecting the HOME position, care must always be taken to ensure that the work safety is selected in such a way that head injuries are avoided.

5.5.3.5 Global transfer points (transfer point) [V_70.00_I6]

Comments must be made in the national language!

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Transfer points are stored as global points in “global_points.dat”.

The movement type E6Axis PTP is used here as a driving command.

```
13 E6Axis PTP <transferpoint_1A> Vel=100 % PDAT1
    Tool[1] Base[0];
```

Transferpoint_1A "-" transferpoint_32A ", are already pre-certified in Global_Points.dat.

If these are not sufficient, additional "Transferpoints_xxx" could be via the E6Axis ILF which

With the basic package, automatically decimated/created.

This E6Axis ILF is approved exclusively for use at transfer points.

Special case for assembly only:

Here, the global transfer points are also permitted as Cartesian points so that the turning points can be pinched in LIN or CIRC.

5.5.3.6 Special points [V_70.00_I5]

Special points are specified by MBAG. These are important positions such as. Pre-adjustment position, mechanical zero etc.

5.5.4 Tool and base name [V_70.00_I5]

Tool and base names are displayed in the inline form of process points. Names shall be assigned for used tools and bases according to the naming conventions

5.5.5 Tool names [V_70.00_I5]

Is to be requested mechanically at a design meeting at the start of the project

Plain text names are written here as examples in German and must be adapted to the specific country!

See Excel table or agreements in the project meetings for translation examples

Composition new:

- Start with function abbreviation
- Second digit always the tool number (not to be confused with tool no.)
- Other points are either progressive change numbers or the component type (country-specific) for grippers

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For details, see GL of Technologies and GL Robotics-Addition (Volume 3 Chapter 40.01.01)

5.5.5.1 Surface Applications [V_70.00_I6]

See additional GL Volume 3-OF: GL-DE_Roboter-Erw.OF

5.5.6 Base name [V_70.00_I6]

Base names are to be assigned according to the convention:

Plain text names are written here as examples in German and must be adapted to the specific country!

See Excel table or agreements in the project meetings for translation examples

All names & comments are to be executed in the national language as 4.4 User programs described in Chapter!

Example: ST010_option_29

ST	010	_option	_29
Base type (see table)	Station number 010	Option: Type, vehicle, component variant, etc.	Option: Progressive change number _1

Table 21 Base names

If the Fahrzeug0 is in the same position of each derivative, use only one base. (Affects bodyshop lines) e.g. ST010

Abbreviation	Meaning
ST	Station
ET	External tool
RR	Robot robot base

Table 22 Base type

Option designations are to be adopted from the design specifications and the file "000_B3.040.01.01_GL_Naming_Robotsoftware_V70.xx" must be observed and do not claim to be complete.

If necessary, they shall be expanded in consultation with MBAG Robot Technology Project Team.

5.5.6.1 External tool [V_70.00_I6]

Example of external tool!

E.g. ET050_KLB_01_1D=> identical to the tools chapter. 5.5.4 Tool and base name [V_70.00_I5]

5.5.7 Tool numbers (TOOL_DATA) [V_70.06]

The vectors of the tool displacements are stored in the $CONFIG.DAT file in the TOOL_DATA [] field from 1 to 128. The name and values are entered via the WoV or BOF. This assignment must not be changed without MBAG's approval.

The lower tool number for the gripper is used without component, e.g.:

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- Tool_Data [10]=GRxx_SW_O
- Tool_Data [11]=GRxx_SW_M

Tool of simulation or tool [128] are only relevant for initial startup and must have the value "0" again after technical acceptance.

To execute all names & comments in national language accordingly!

Direct assignments to TOOL_DATA in the program run are not permissible.

Tool no.	Tool name	Meaning (only shown in German)
TOOL_DATA [1]		Tool 1 (e.g. Pliers, adhesive head, etc..)
TOOL_DATA [2]		Tool 2
TOOL 3.. 9		Tool 3 to 9
TOOL_DATA [10]		Gripper 1 (without component)
TOOL_DATA [11]		Gripper 1 (with component)
TOOL_DATA [12]		Gripper 1 (veh 0)
TOOL 13 ... 19		Gripper 1 auxiliary tool
TOOL 20...29		Gripper 2 (number x1 to x9)
TOOL 30 ... 39		Gripper 3 (number x1 to x9)
TOOL 40 ... 49		Gripper 4 (number x1 to x9)
TOOL 50 ... 58		Auxiliary tools for camera position in technology package DAI_SEN
TOOL_DATA [59]		Calculation tool for 3D stacking
TOOL_DATA [60]	Exchangeable disk	Tool change plate (see GL Tool Change)
TOOL_DATA [61]	Exchangeable Disk_Load	Tool change plate with other load data
TOOL_DATA [62]	Safe ref	Safe Ref (tool for safe-OP reference tuning fork Load data M from the current tool that is mounted or docked)
TOOL_DATA [63]	Commissioning	Reserved for tip tool for base measurement
TOOL_DATA [64]	Zero	Zero data (➔load data M=0kg)
TOOL 65 - 68		Reserved for standardization
TOOL_DATA [69]	Console	Auxiliary tool of the console data for calculation tool tool e.g. For spot welding guns-measurement report
TOOL 70 - 99		Reserve
TOOL 100 - 109		Charged tools for correction systems e.g. TCP-3D rail. Or other corrections
TOOL 110 - 119		Reserve
TOOL 120 - 128		Tools for simulation

Table 23 TOOL_DATA Number assignment

5.5.8 Base numbers (BASE_DATA) [V_70.02]

The vectors of the base displacements are stored in the file $CONFIG.DAT in the field BASE_DATA [] from 1 to 64. The name and values are entered via the BOF. This assignment must not be changed without the approval of the Robotics project team Mercedes-Benz.

Explanation:

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- Veh. Coordinates=> Vehicle-related basis that is calibrated from the design with a measuring system into the system
- External tools=> External tools-Tools
- External base=> Coordinates the measured base with a measurement system in the system were calibrated e.g. Stowage boxes, load containers, etc.
- Master base measurement and associated calculated base=> reserved for measuring tasks where base is calculated. Master base is the calibrated basis and the calculated base is used for storage
- Calculated base displacement (Base_DATA [60-63])=> is used for calculating the measurement data for the motion of the robot!
- Calculated base displacements with Masterbases (e.g. Measurement systems)=> serves to create masterbases that should not be overwritten
- Direct assignments to BASE_DATA in the program run are not permissible.

To execute all names & comments in national language accordingly!

Base no.	Base name	Meaning (only shown in German)
BASE_DATA [1]		Fzg.Koordinaten 1
BASE_DATA [2]		Fzg.Koordinaten 2
BASE_DATA [3]		Fzg.Koordinaten 3
BASE_DATA [4]		Fzg.Koordinaten 4
BASE_DATA [5]		Fzg.Koordinaten 5
BASE_DATA [6]		Fzg.Koordinaten 6
BASE_DATA [7]		Fzg.Koordinaten 7
BASE_DATA [8]		Fzg.Koordinaten 8
BASE_DATA [9]		Fzg.Koordinaten 9
BASE_DATA [10] - [19]		External tool 1-10 and Roboteam
BASE_DATA [20] - [29]		External base 1-10 e.g. Tables, stowage boxes, load containers, etc.
BASE_DATA [30] - [59]		Master base measurement and associated calculated base
BASE_DATA [60]	CalcBaseMovement1	Calculated base displacement 1 (short memory)
BASE_DATA [61]	CalcBaseMovement2	Calculated base displacement 2 (short memory)
BASE_DATA [62]	CalcBaseMovement3	Calculated base displacement 3 (short memory)
BASE_DATA [63]	CalcBaseMovement4	Calculated base displacement 4 (short memory)
BASE_DATA [64]	Zero	Zero data
BASE_DATA [65] - [90]		Fzg.Koordinaten 10-35
BASE_DATA [91] - [109]		External base 11-20 e.g. Tables, stowage boxes, load containers, etc.
BASE_DATA [110] - [119]		Base for camera position for technology package DAI_SEN
BASE_DATA [120] - [128]		Reserve

Table 24 BASE_DATA Number assignment

5.5.9 Tool/Base Examples at KUKA.Roboteam [V_70.00_I6]

When using KUKA.Roboteam, the tools/bases must be linked in the "GeoLink" command. This is why the specified robot identifies this as $tool "_LK" and the following robot number is entered as the base for the following robot Ex. "_R100".

Additional tool for predetermined robots with command "GeoLink ":

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- Tool 13 Tool with GeoLink Roboteam designation: GRH_01_xxxx_LK

Additional base for the following robot with command "GeoLink ":

Robot name with which it is coupled

- Base 13 Base GeoLink for Device Robot Designation: GRH_01_xxxx_RB100

5.5.10 Correction via external measurement equipment [V_70.00_I6]

The existing $Tools or $Base can be changed via external measurement equipment on the robot to make adjustments.

Basically, the original tool or base must never be overwritten, but the original must always be described with the accounting value of the measuring system to a new tool or base!

The following boundary conditions must be observed for this

5.5.10.1 Correction $Tools [V_70.00_I6]

This function shall never be used for gripper applications, but only via approved technology package functions! It is absolutely essential that the original tool is never overwritten, but is used with the tools for invoicing ($Tool 70-79)

Example of arc welding via the TCP-3D measurement determines the wire center point for optimum rail guidance or for corrections from other measuring systems in the surface or assembly (special project approval required).

5.5.10.2 Position correction $Base [V_70.00_I6]

This function is for all robots where the robot has to be corrected.

Examples are corrections to include parts in the correct position or correctly install components in another component/body (example: Glass pane assembly)

5.6 I/O programming

5.6.1 Long text names [V_70.00_I6]

To execute all names & comments in national language accordingly!

The designations for inputs and outputs shall be available as long text names in the BOF as per the PLC designation system. The entry must in a text file the name of the station with the ending txt, (e.g. SW11_060RB200.txt). The file must contain all long texts of the standard technologies used and the system-specific long texts.

All long texts are edited and saved in the WorkVisual in the project. The country-specific long texts must be entered there.

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For the long texts used in the technology package, these are automatically imported into the WorkVisual project during import.

Make sure that designations are changed to the WorkVisual project in the BOF on the SmartPad if designations are changed on site.

For systems or station-specific signals such as. Locking signals, the naming convention consists of the PLC signal name (12 characters), and a comment separated by a colon.

Example:

Anti-collision with 070RB200

$IN [1]=> "Clamper open GS (040GRP301-QM1MM1BG21:4)"

=> as before

Long texts may be a maximum of 90 characters long and contain no umlauts.

Plain text names are written here as examples in German and must be adapted to the specific country!

5.6.2 IO-Konfiguration der Feldbusse [V_70.00_I6]

5.6.2.1 EA Tables Profinet [V_70.00_I6]

There is only one E/A field bus list

<u>Basic list</u>

Documents are in Volume 3\ 040.\ 01. Guideline for Standard Robot\ 03. Mercedes-Benz E-table stored.

This describes the basic Profinet device assignments as well as communication with the superordinate PLC.

This basic list is binding for all shops and projects, as this also includes the standard modules used in the PLC

5.6.2.2 Profinet naming [V_70.03]

⇨ See also Chapter 1.9 Naming of robot system [V_70.02].

The EA field bus list shows that the KF designation is identical to the user ID in the configuration in the robot. This is absolutely necessary because the user ID is identical to the Profinet subprojects for importing into the project contained in the technology packages. This is then used for the Udpate mechanism as well as for the off/docking docking of Profinet nodes via the tool change technology packages.

None of this may be deviated from! Excaption:

- IMT-Perceptron: Device names always kf190 and IDs according to the Profinet devices from the tech package.

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Example:

Part import technology package:

Profinet node:

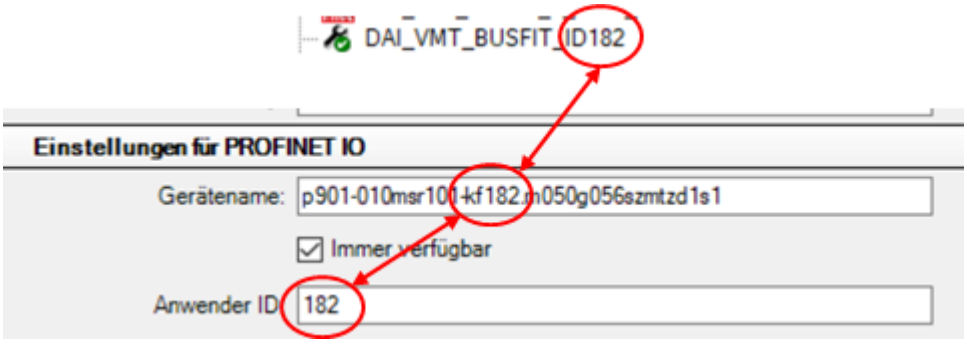

Figure 33 Profinet node designation

5.6.2.3 Profinet configuration [V_70.02]

The KUKA documentation always applies to the Profinet configuration.

The cycle time settings of the Profinet nodes shall be set in accordance with Profinet specification from Volume 3 Chapter 26. Individual technologies have smaller, as well as switches, even larger cycle times. Cycle times below 2ms are not permitted!

In the EA table for PN robot Volume 3 Chapter 40.01.03, the PN device numbers are stored as well as the special cycle times.

<u>The following functions are not permitted:</u>

- Profinet function "Shared Device" is not permitted=> Exception IMT – Perceptron
- IO-Link function is not permitted

5.6.2.4 Profinet Topology [V_70.05]

Topology must be created in the WoV project and saved to the control system so that the partner port (offline) and partner port (online) are identical!

Attention, is not possible if the robot has no PN nodes or is the IMT-Perceptron technology!

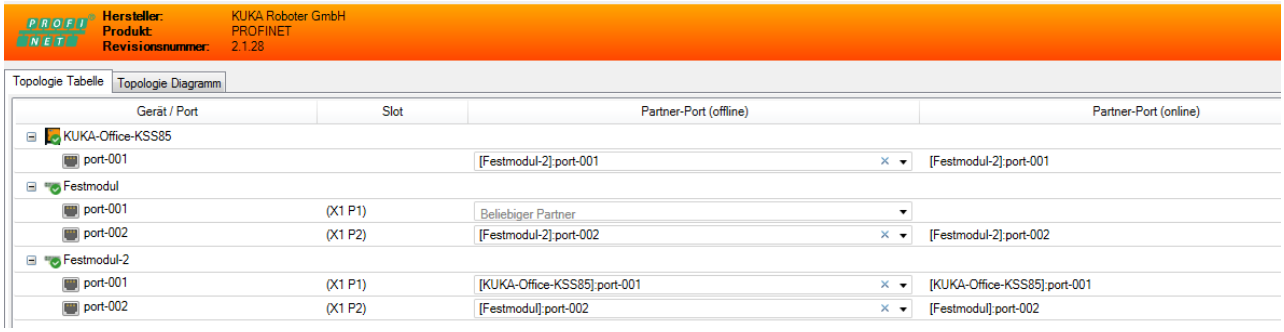

Figure 34 Profinet topology

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5.6.3 Signal states after robot stop [V_70.00_I5]

Currently no longer planned. If this function is required, this must be discussed with the Robotics Project Team.

5.6.4 Basic state of I/O 's (user_SPS_INI.src) [V_70.00_I5]

The user_sps_ini.src file is automatically executed in the "External" operating mode when the robot is in home position by the submitter. This makes it possible to set the outputs, variables and interchangeable tools to a defined state so that a restart of the robot without manual intervention is also possible even after a cold start or E/A conf.

5.6.5 Diagnosis of I /O s and Variables [V_70.00_I5]

The use of "configMon.ini" may therefore currently only be carried out for displaying project-specific variables that are NOT covered by the standards.

The KRC5 controller offers the possibility to display variables for diagnostic purposes in a dedicated window. Which inputs and outputs are to be displayed must be configured in the file "ConfigMon.INI ". Project planning takes place via the configuration interface.

If necessary, important variables must be projected into the Configmon.ini at runtime in coordination with the system operator in order to be able to monitor these during the program sequence.

5.6.6 Additional function "Plus communication" [V_70.00_I5]

In the case of positive communication, the quality information (NOK/OK) and point number are entered in a table in the corresponding technology packages after the process point. During the main program end/component, the collected table information is then transferred to the PLC via the acyclic service from Profinet. The content is then linked to all other information in the PLC and sent to a higher-level system-specific.

See also documentation from base ect, as well as for PLC modules in Chapter 26

5.6.7 Additional function "ext. joining point number Communication "[V_70.00_I5]

RDS communication refers to the extended joining point number, which requires individual corresponding technologies for processing RDS data.

The data is provided in the PLC in the "data channel" and made available to the robot on the interface in the data channel. This is then sent internally by robot 1:1 to all corresponding technologies without making changes.

See also documentation from base ect, as well as for PLC modules in Chapter 26

5.6.8 Additional function "Dyn-PLC" for technology PLC [V_70.00_I6]

Replaces the former Visu signals and EA interface for bits for exchange between technology and PLC via the robot!

The assignment takes place via the corresponding technology packages that use this function.

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The user must then select the order in the WorkVisul in the DynPLC editor.

Note that only a maximum of 5 DynPLC areas can be selected!

See also documentation from base ect, as well as for PLC modules in Chapter 26

5.7 Programming IoT capability

5.7.1 IoT data system [V_70.00]

With the basic installation of the Mercedes-Benz AG MQTT driver, data from the system including the messages and log content are sent away cyclically.
The line builder doesn't have to parameterize anything here!

5.7.2 IoT Data Process/User-Specific [V_70.00_I6]

To execute all names & comments in national language accordingly!

In order to be able to submit additional information in the process flow outside of technology packages, the system builder is required to incorporate this accordingly. These should serve for better preventive diagnosis as well as for vehicle-related diagnosis and information. The exact values in the individual shops of concerns are regulated in Volume 2 accordingly in the project.

To send the respective information from the application program, a simplified form is then provided by Mercedes-Benz.

5.8 Program archiving [V_70.03]

The contractor is always responsible for archiving the robot programs. The contractor shall perform manual archiving regularly and after changes. File → Archive → Backup Manager → Backup

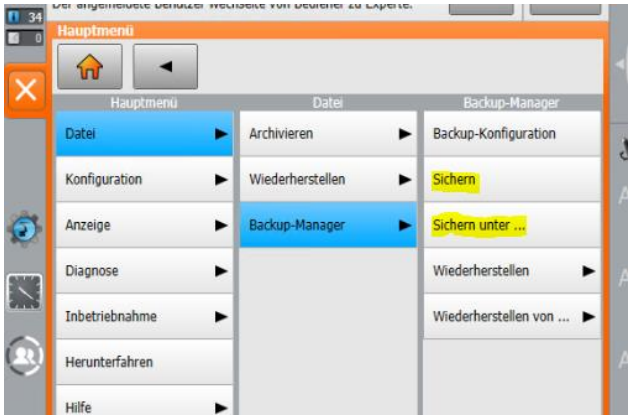

Figure 35 Program archiving

During commissioning, the robots are connected to an archiving server via the plant network.

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It must be ensured that created archives are also accessible for MBAG as long as the archive server is not yet available.

5.9 RoboTeam Robot [V_70.00]

For the cooperating robots, the directory in Volume 3 contains Robo team folders, provided that this is enabled in the project. Standard is not planned.

Robo team may only be used in Integra phase 7 for the "load sharing" function! However, special approval is required from the Mercedes-Benz Robot Project Team!

Exception is only permitted with written approval from the Mercedes-Benz Robot Project Team if the RoboTeam network uses the load sharing function for transport to robots. No further technology packages with processes may be used!

Robo team currently not planned, therefore only against special approval from robotics project team!

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6 Program documentation and acceptance [V_70.06]

Attention, all documentation relevant data are in the structure in Volume 3 Chapter 002.04.32!

All robots whose documentation is no longer correct after integration/conversion have to be adapted. This also applies to robots that were not directly affected by the conversion, but whose change concept was influenced by the conversion, for example.

In which form the documents are submitted, refer to the specifications of the documentation in Volume 3 Chapter 002..

6.1 General [V_70.02]

The documentation described here shall generally be created for each robot. If documents are not created per robot, this is the case in the folder.

Excerpt from the overview of the folder Volume 3 Chapter 002.04.32 Robot

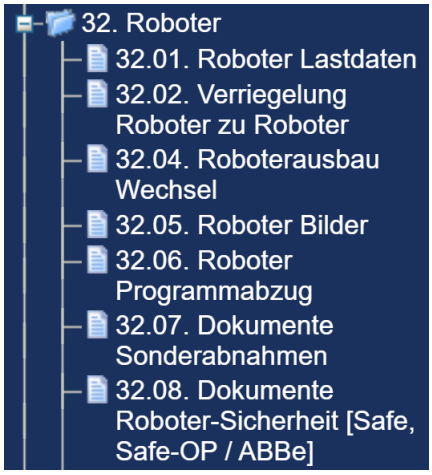

Figure 36 Directory structure of documentation in ECM

6.2 Folder content B3.002.04.32 [V_70.00_I5]

All documents must always be delivered in the current Office format.

The specifications from the overall documentation specifications apply to PDF files.

6.2.1 32.01 Robot Load Data [V_70.04]

The Kuka-Load-Online tool should be used to document the load data.

Content:

A PowerPoint or PDF file must be submitted for each robot and tool or gripper with the heaviest component.

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1. Design data of the tool including component and additional load data (design release)
2. Real data (KukaLoadProject.xxxxxx.xml) after technical acceptance

6.2.2 32.02 Robot Lock to Robot [V_70.00_I5]

Here the robot-robot interlocks are divided into meaningful areas, whereby the relationships between the individual robots must be recognizable.

Content can be submitted either in Word format according to the template or as PowerPoint from the simulation.

Content:

Per connected robot area per system

⇨ Delivered as Word or PowerPoint files

6.2.3 32.03 Robot info sheet [V_70.02]

Not applicable in integra phase 7

6.2.4 32.04 Robot Removal Change [V_70.00_I6]

For each robot, the removal concepts are shown that are possible with the equipment of the location=> see Volume 2 Project Specifications

Content:

Expansion concept per robot

6.2.5 32.05 Robot Images [V_70.00_I5]

Each robot displays the individual images here.

It is important that the images are clearly recognizable and not too dark.

The images must have a resolution of 800x600.

Name of the images so that later also in robot folder can be copied:

VB21_030RB_100

Content:

Per robot:

- Adjustment positions per tool from both side views. The hose package position and tennis rackets must be clearly recognizable!

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- Tool changer-robot side (close-up of cable duct)
- Pictures of special hose packages that are not in the supply process or have special challenges for routing. E.g. BestFit

6.2.6 32.06 Robot Program Deduction [V_70.00_I5]

A memory withdrawal is set for each robot.

The corresponding KRC5 data must be documented in the "KRC" directory. Usually, the user interface uses "Archive everything" to create a deduction that is then transferred according to the specifications of documentation GL-DE B3/Chapter 002..

Content:

One storage deduction per robot

Agreement with the documentation team Project whether data from the robot archiving server can be copied directly! To be clarified for specific locations.

6.2.7 32.07 Acceptance Documents Special Acceptance [V_70.03]

Special acceptance processes are set for each robot, which are required for additional documentation.

6.2.7.1 Hose package-documentation process [V_70.03]

The contractor can have this weight removed by the hose package manufacturer according to the manufacturer's specifications. In principle, the contractor gives us the warranty according to the provision scope (Volume 2/Robot).

For hose packages that are NOT provided by Mercedes-Benz, the system supplier shall still present an acceptance document!

Queries about the process for the robotics project group.

The documentation shall be submitted in accordance with the documentation GL-DE B3/Chapter 002 specifications.

Content:

Per robot

- Special hose packages with acceptance by hose package manufacturer

6.2.7.2 Linear unit-documentation process [V_70.03]

Acceptance of KUKA linear unit by Kuka => see also Chapter 1.5.4.1 Acceptance of Linear Unit [V_70.06].

Content:

Per robot

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- Checklist with measurement protocol

6.2.8 32.08 Robot Safety Acceptance Test Report [V_70.00_I6]

The respective safety acceptance protocols are set for each robot.

6.2.8.1 32-08 Robot acceptance without active safe-OP [V_70.00_I6]

Content:

Per robot, if only standard safe:

- Without iPortal CRC monitor:
  - Only WorkVisual documentation printout with signature (scanned in)
    - *Not applicable if iCRC system is used. See agreement in the robot project rekoplog!*
- With iPortal CRC monitor:
  - No documents

6.2.8.2 32-08 Robot acceptance with Active Safe-OP [V_70.00_I6]

Content:

Per robot, if safe-OP function active:

- Document: Acceptance safe OP presentation with the data from the offline simulation (OLP-Delmia macro) OG other sources from which the Cellspace, rooms and reference position are shown in the layout of the system and the ball sets of the tool are based on the tool CAD drawing
- Document: Acceptance checklist filled out per robot
- Document scanned with signatures:
  - Acceptance checklist for cover sheet with signatures (scanned in) or acceptance checklist sum sheet with signature (scanned in)
- Document scanned in: CV must be scanned with signatures, original remains in robot cabinet
- Current WOV project with activated safe-OP function

There are separate documents for LBR robots

6.2.8.3 32-08 Robot acceptance with safe axle limitation [V_70.00_I6]

It is important that the axle range limitation is validated by a safe-OP zone-specific axis in order to prevent destruction of the robot outside of the hazard, e.g. In the T1!

Content:

Per robot, if installed with safe axle range limits:

- Acceptance document as per specification

6.3 Acceptance of MRC Applications [V_70.00_I6]

In the case of robots working in MRK mode, the results of the tests on site are documented in this chapter to the biomechanical limit values in accordance with the specification ISO TS 15066.

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6.4 Acceptance [V_70.00]

Acceptance criteria are contained in the checklists in Volume 3 Chapter 009 and are 6.4.1 Acceptance Tool [V_70.00_I6] carried out with the acceptance tool from Chapter.

Here only small extract regarding the control as a reference point for handover:

- Are all not required and self created programs (test programs) deleted
- No external programs or files under Windows
- Technology packages in original state
- Long texts are included in the current workvisual submission project
- Old work-visual projects are deleted

6.4.1 Acceptance Tool [V_70.00_I6]

The WoV test routines are made available by MBAG in the project to the plant manufacturer and internally at MBAG it is used to check the tracking points from the Coman.

Information on this in the robot regular communication.

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7 Technology packages

7.1 General [V_70.00_I6]

Robot technology packages are software modules of the robot control system that define the process and programming of the respective process. By default, all technology packages described here are made available by MBAG or the robot system supplier via installation software in WorkVisual. The SRC files of the technology packages may not be changed; only the planned parameterizations in the DAT files may be adapted.

If further technology packages are required in the project that do not match the packages provided by Mercedes-Benz, this must be clarified with the Mercedes-Benz robot project team. Mercedes-Benz is committed to either modifying an existing technology package or developing a new one. If this is rejected by Mercedes-Benz, the technology package including documentation is always the contractor's scope of delivery. For the creation of additional technology packages, the contractor shall adhere to the specifications for creating technology packages from MBAG. An example technology package is provided by MBAG. Coordination with the Mercedes-Benz Robot Project Team is mandatory.

The implementation of a technology package after the installation of the associated hardware and software is divided into:

1. Parameterization and initial startup
2. Programming
3. Function test
4. Optimization
5. Acceptance by the Robot Project Team of Mercedes-Benz AG

In addition to the robot guideline contained here, the process descriptions belonging to the respective technology must also be observed. For complex technologies (e.g. Laser welding/soldering, Robscan, bonding, seam sealing, ...) may also be necessary to visit additional training at the technology manufacturer and the robot supplier. The contractor shall only use personnel for programming the robots that possess the necessary knowledge about the respective technology.
The knowledge is to MBAG by supporting documents (e.g. Training confirmation).

7.2 Technology packages standard [V_70.01]

All current technology packages are only stored in the provided robotics software ECM project of the respective project.

There are technology packages that apply equally to all plants, but also documents that apply only to one plant.

Some technology folders are described separately due to the special content.

7.2.1 Documentation of technology package [V_70.01]

All documentation for the technology package is available in the iKop file.

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These are either accessible via the WorkVisual (WoV) from KUKA or by unpacking a Zip file!

Example here:

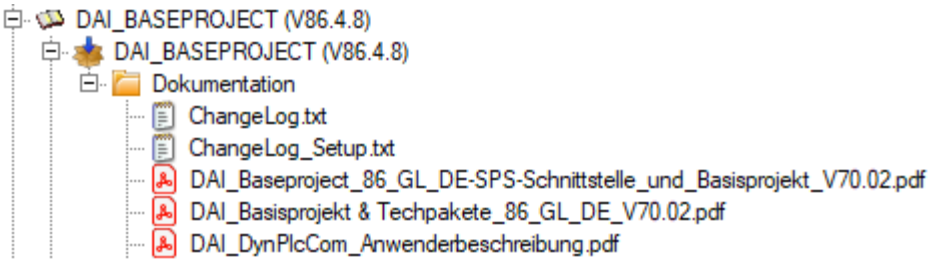

Figure 37 Unpack technology package documents

7.2.2 Parameterization and initial startup [V_70.00_I5]

The E/A assignments of the field bus subscribers used on the robot are mandatory prerequisites for commissioning. MBAG assigns certain address ranges and participants to all field bus subscribers that must be configured by the contractor as specified (field bus list). Depending on the field bus system used, the costs for the configuration are different and shall be taken into account by the contractor.

Technology packages have a parameterization part in which the signal assignments as well as the corresponding options can be programmed depending on the package. When installing via the MBAG Work Visual Option Editor, a standard parameterization is included that has to be checked by the contractor and adapted to its specific situation if necessary. No changes may be made except in the parameterization part.

Following the bus configuration and parameterization of the technology packages, an E/A function test of the safety-related signals as well as the controller and device side of the fieldbus system must be performed.

7.2.3 Programming [V_70.00_I5]

Programming is divided into motion and process programming. A user interface is available to this end, so-called inline forms with which the programmer can enter the associated parameters.

The motion guidance shall always be programmed with approximate positioning with the exception of process and holding points

7.2.4 Function test [V_70.00_I5]

The function test includes the run-off of all programs in the intended operating mode.

7.2.5 Optimization [V_70.00_I5]

The optimization takes place after a successful function test and includes the following:

- Movement optimization
  - It shall be ensured in this case that sufficient safety distance to interference contours of the system is maintained.
- Process optimization
  - Includes adaptation of parameters of technology control and robot system (e.g. Welding parameters such as retention time).

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- Process optimization (locks, maintenance statuses)
  - Is to be performed in external operation. This is intended to test system and robot locking scenarios, maintenance statuses, etc.
- Hose package guide
  - Ensure that the hose package is routed without kinks during the entire movement. If necessary, additional joints or guides shall be attached in consultation with the Mercedes-Benz Robot Project Team.

7.2.6 IP address distribution on robot [V_70.00_I5]

The IP addresses of the KRC5 can be divided into 3 areas.

7.2.6.1 ProfiNET [V_70.00_I5]

The IP address of the Profinet stack, and thus also the IP address of Windows (only one IP address) for Mercedes-Benz, is covered and assigned by the control technology process.

7.2.6.2 IP direct assignment [V_70.03]

KRC5 is not possible for the RSI/EthernetKRL interface to the external system.

The RSI/EthernetKRL interface thus hangs physically in the hall and Profinet network of the system.

Thus, administration of these addresses is necessary to avoid address conflicts.

Each technology has its own area of once specified.

Technology	Robot area	Participant area	Note
RobScan	192.167.[Zelle].[11 … 19]	192.167.[Zelle].[31 … 39]	Roboter zusätzliche IP 192.
RobiScan	192.167.[Zelle].[11 … 19]	192.167.[Zelle].[31 … 39]	Roboter zusätzliche IP 192.

Table 25 Technologies with special IP areas

It makes sense to use a compliant ProfiNet interface to avoid proprietary interfaces such as RSI or EthernetKRL.

7.2.6.3 Internal/blocked IP addresses [V_70.00_I5]

The following IP address ranges are used internally for the KR C4 and may therefore not be used in the vicinity of the KR C4

- ⇨ 172.17 X X.X
- ⇨ 172.16 X X.X
- ⇨ 192.168.0. X

7.3 Basic function packages [V_70.00_I5]

7.3.1 PLC automatic mode [V_70.00_I5]

This document describes the E/A interface for the robot PLC.

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The content refers here to the function module of the PLC, so that the robot can execute the standard functions, e.g. Program selection, ghost, etc.

7.3.2 Field bus [V_70.00_I5]

The respective guideline is stored in the field bus folder and the E/A configurations for the projects.

7.4 Messages [V_70.00_I6]

Own user information messages, acknowledgment messages and dialog questions that are not included in the technology packages may only be sent via the functions in the "dai_msg_library.src ".

User texts are to be executed in the national language.

As of Integra phase 7, all messages are also loaded into the cloud in accordance with IoT capability. Therefore, messages outside the system and technology packages are also absolutely necessary. See Chapter 5.7 Programming IoT capability.

7.5 Example/and utilities [V_70.02]

Provision of sample programs or templates in ECM robotics projects only for programs included in the base project. Examples of the technology packages can be taken directly from the respective technology package (via Work Visual Export function).

These are then made available in the corresponding technology packages (Kop files).

Errors in the example programs shall be cleaned up by the contractor during copying or changes in accordance with agreements in the robot project regular communication.

7.6 Program Packages at Window Level [V_ 70.00_I5]

If your own programs, drivers or DLLs have to be installed at the Windows level (e.g. For measuring robots), this can only be done with written approval from KUKA and MBAG.

Details must be clarified with the Mercedes-Benz Robot Project Team.

7.7 Line builder creates a technology package code [V_70.00]

If there is no Integra standard tech package for a technology, this must be provided by the line builder. The following options are permitted for this:

- Tech package with SRC/DAT-Code
- Tech package with Kuka KOP mechanism

Exclusion: Tech package name must not be called "DAI"!

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7.7.1 Technology package with only SRC/DAT code [V_70.00]

In this case, no inline forms or display elements (HMI Easy) may be used. Here the SRC/Dat file must be stored in a subfolder in the "Program folder" and not in the TP folder.

7.7.2 Technology package with the KOP mechanism [V_70.00]

In this case, inline forms and display elements (HMI-easy) can be used, as well as I/O module links. Profinet participants must be preconfigured as sub-projects including long texts in the kop file.

The tech package must be created as a kop file using the KUKA option package editor (OPE). The processes for reinstalling, updating, etc. must be ensured.

The tech package must be placed in a subfolder in the "TP folder".

7.7.3 EA addresses [V_70.04]

The I/O area from 4705 - 5056 (44 bytes) can be used by special applications. If the area is not sufficient, this must be agreed with the robotics project team.

For the technology interface to the PLC, the Dyn-PLC interface is to be used; The ID required for this can be requested from the standardization for PLC. There is a limit of 40 bits of input/output per ID.

7.7.4 User areas [V_70.03]

The following areas are available for special applications:

- Interupt → 101-110
- Cycleflag → 101-110
- Timer → 101-110

7.7.5 Access to MBAG-Baseproject functions [V_70.04]

All basic project functions may generally be used, see documentation of base project.

If a technology requires a submit module, these must be integrated per file manipulation in the Tech-Loop (IDs "101-110") (file: DAI_Tech_loop.src)

The "Plus data" function may be used, but the plus ID must then be requested from the production planner via the project.

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